

COVER SHEET FOR PROPOSAL TO THE NATIONAL SCIENCE FOUNDATION

PROGRAM ANNOUNCEMENT/SOLICITATION NO./CLOSING DATE/if not in response to a program announcement/solicitation enter NSF 04-23					FOR NSF USE ONLY	
NSF 00-144					NSF PROPOSAL NUMBER	
FOR CONSIDERATION BY NSF ORGANIZATION UNIT(S) (Indicate the most specific unit known, i.e. program, division, etc.)					0617136	
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DATE RECEIVED	NUMBER OF COPIES	DIVISION ASSIGNED	FUND CODE	DUNS# (Data Universal Numbering System)	FILE LOCATION	
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Middlebury College			Middlebury College			
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0036913000			Middlebury, VT. 057536000			
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IS Awardee ORGANIZATION (Check All That Apply) (See GPG II.C For Definitions)			☐ SMALL BUSINESS ☐ FOR-PROFIT ORGANIZATION		☐ MINORITY BUSINESS ☐ WOMAN-OWNED BUSINESS	
			☐ IF THIS IS A PRELIMINARY PROPOSAL THEN CHECK HERE			
TITLE OF PROPOSED PROJECT RUI: Memory for perceiving information that was never experienced						
REQUESTED AMOUNT \$ 175,169		PROPOSED DURATION (1-60 MONTHS) 36 months		REQUESTED STARTING DATE 09/01/06		SHOW RELATED PRELIMINARY PROPOSAL NO. IF APPLICABLE
CHECK APPROPRIATE BOX(ES) IF THIS PROPOSAL INCLUDES ANY OF THE ITEMS LISTED BELOW						
☐ BEGINNING INVESTIGATOR (GPG I.A) ☒ HUMAN SUBJECTS (GPG II.D.6) ☐ DISCLOSURE OF LOBBYING ACTIVITIES (GPG II.C) Exemption Subsection _____ or IRB App. Date Pending ☐ PROPRIETARY & PRIVILEGED INFORMATION (GPG I.B, II.C.1.d) ☐ INTERNATIONAL COOPERATIVE ACTIVITIES: COUNTRY/COUNTRIES INVOLVED (GPG II.C.2.j) ☐ HISTORIC PLACES (GPG II.C.2.j) ☐ SMALL GRANT FOR EXPLOR. RESEARCH (SGER) (GPG II.D.1) ☐ VERTEBRATE ANIMALS (GPG II.D.5) IACUC App. Date _____ ☐ HIGH RESOLUTION GRAPHICS/OTHER GRAPHICS WHERE EXACT COLOR REPRESENTATION IS REQUIRED FOR PROPER INTERPRETATION (GPG I.G.1)						
PI/PD DEPARTMENT Psychology			PI/PD POSTAL ADDRESS Department of Psychology			
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CERTIFICATION PAGE

Certification for Authorized Organizational Representative or Individual Applicant:

By signing and submitting this proposal, the individual applicant or the authorized official of the applicant institution is: (1) certifying that statements made herein are true and complete to the best of his/her knowledge; and (2) agreeing to accept the obligation to comply with NSF award terms and conditions if an award is made as a result of this application. Further, the applicant is hereby providing certifications regarding debarment and suspension, drug-free workplace, and lobbying activities (see below), as set forth in Grant Proposal Guide (GPG), NSF 04-23. Willful provision of false information in this application and its supporting documents or in reports required under an ensuing award is a criminal offense (U. S. Code, Title 18, Section 1001).

In addition, if the applicant institution employs more than fifty persons, the authorized official of the applicant institution is certifying that the institution has implemented a written and enforced conflict of interest policy that is consistent with the provisions of Grant Policy Manual Section 510; that to the best of his/her knowledge, all financial disclosures required by that conflict of interest policy have been made; and that all identified conflicts of interest will have been satisfactorily managed, reduced or eliminated prior to the institution's expenditure of any funds under the award, in accordance with the institution's conflict of interest policy. Conflicts which cannot be satisfactorily managed, reduced or eliminated must be disclosed to NSF.

Drug Free Work Place Certification

By electronically signing the NSF Proposal Cover Sheet, the Authorized Organizational Representative or Individual Applicant is providing the Drug Free Work Place Certification contained in Appendix C of the Grant Proposal Guide.

Debarment and Suspension Certification

(If answer "yes", please provide explanation.)

Is the organization or its principals presently debarred, suspended, proposed for debarment, declared ineligible, or voluntarily excluded from covered transactions by any Federal department or agency?

Yes ☐

No ☒

By electronically signing the NSF Proposal Cover Sheet, the Authorized Organizational Representative or Individual Applicant is providing the Debarment and Suspension Certification contained in Appendix D of the Grant Proposal Guide.

Certification Regarding Lobbying

This certification is required for an award of a Federal contract, grant, or cooperative agreement exceeding \$100,000 and for an award of a Federal loan or a commitment providing for the United States to insure or guarantee a loan exceeding \$150,000.

Certification for Contracts, Grants, Loans and Cooperative Agreements

The undersigned certifies, to the best of his or her knowledge and belief, that:

(1) No federal appropriated funds have been paid or will be paid, by or on behalf of the undersigned, to any person for influencing or attempting to influence an officer or employee of any agency, a Member of Congress, an officer or employee of Congress, or an employee of a Member of Congress in connection with the awarding of any federal contract, the making of any Federal grant, the making of any Federal loan, the entering into of any cooperative agreement, and the extension, continuation, renewal, amendment, or modification of any Federal contract, grant, loan, or cooperative agreement.

(2) If any funds other than Federal appropriated funds have been paid or will be paid to any person for influencing or attempting to influence an officer or employee of any agency, a Member of Congress, an officer or employee of Congress, or an employee of a Member of Congress in connection with this Federal contract, grant, loan, or cooperative agreement, the undersigned shall complete and submit Standard Form-LLL, "Disclosure of Lobbying Activities," in accordance with its instructions.

(3) The undersigned shall require that the language of this certification be included in the award documents for all subawards at all tiers including subcontracts, subgrants, and contracts under grants, loans, and cooperative agreements and that all subrecipients shall certify and disclose accordingly.

This certification is a material representation of fact upon which reliance was placed when this transaction was made or entered into. Submission of this certification is a prerequisite for making or entering into this transaction imposed by section 1352, Title 31, U.S. Code. Any person who fails to file the required certification shall be subject to a civil penalty of not less than \$10,000 and not more than \$100,000 for each such failure.

AUTHORIZED ORGANIZATIONAL REPRESENTATIVE		SIGNATURE		DATE	
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Project Summary for RUI: Memory for perceiving information that was never experienced

The research proposed in this grant application seeks to understand the theoretical basis for the recollection of perceptual details in the absence of perceptual experience. Specifically, the proposed research will evaluate one central question: **are the representations underlying false memories associated with perceptual features that were present during encoding of related information, despite the fact that those memories were never perceived?** I propose to study this question via the use of a basic memory paradigm where subjects are presented with a series of items (e.g., *saddle, pony, mare, gallop, colt, stallion, rodeo, stable, trot, and unicorn*; referred to as *themes* below) that are all associatively related to a single word that is never presented during encoding (e.g., *horse*), referred to as the *lure item*. This paradigm allows one to study false memories for an event that was never perceptually experienced by participants in a basic memory paradigm where experimental parameters can be systematically manipulated. Prior research has demonstrated that people strongly believe that lure items were encoded, as evidenced by high lure error rates and subjective judgments that they can recollect details of lure item presentation. In order to address the question of whether or not the representations underlying false memories are associated with perceptual features present at encoding, the research proposed here makes use of the method of converging operations. In particular, I will use two memory tasks that range widely in their underlying retrieval demands: recognition memory tasks and perceptual implicit memory tasks. The **first hypothesis** tested is that lure items will be associated with perceptual encoding context, which will be revealed on both of these memory tasks, because both tasks have been demonstrated to be sensitive to the overlap between the perceptual features encoded at study and those presented at retrieval.

The proposed research also seeks to assess the extent to which associations between lure item representations and perceptual features encountered at encoding are strengthened by particular encoding circumstances. Prominent theories of memory errors suggest that the development of associations between lure items and perceptual features present during encoding is critically dependent on the extent to which encoding circumstances activate the memory representations of lure items. Thus, despite the fact that lure items are by definition never perceived, it is possible that their representations in memory still become highly active when a number of associatively-related items are perceived during encoding. This high level of activation in turn may inspire the formation of associations between lure item representations and features of the encoding context, such as visual or auditory perceptual records. Therefore, the **second hypothesis** tested is that when variables are manipulated that increase the activation of lure representations (e.g., the strength of the association between studied and lure items, the number of items related to a lure that are encoded, and when participants attend to the similarities among items within a theme), the association between a lure's representation and perceptual encoding context will be stronger.

Intellectual Merit: The results of these studies will have potential implications for understanding the nature of the representations people rely on when they form and retrieve memories for events that they never perceptually experienced. To the extent that the results of these studies suggest that lure representations are associated with perceptual encoding context, they will provide an explanation for why people strongly believe that lure items were actually experienced and claim to be able to recollect the occurrence of the lure item during encoding. Specifically, such results would suggest that people retrieve actual perceptual records associated with lure representations.

Broader Impacts: As part of this research, a minimum of three undergraduate students per year will be employed as research assistants. These students will be given intensive first-hand experience in all aspects of research. Further, student researchers will be involved in the process of dissemination via both publication and conference attendance. Finally, these studies investigate the underlying bases of false memories. If the results suggest that people retrieve authentic perceptual information in conjunction with lure representations, those results would provide one potential basis for the often detail rich, but erroneous memories that people produce in real-world situations such as eyewitness memory accounts.

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*Proposers may select any numbering mechanism for the proposal. The entire proposal however, must be paginated. Complete both columns only if the proposal is numbered consecutively.

Results from prior NSF support

The PI was co-PI for a 2005 summer Research Experience for Undergraduates program (NSF-REU # 0452869, entitled “Research in Psychology: Discovering the Process; Award Amount \$60,032) that supported 9 students as summer research interns to work with six Middlebury College Psychology faculty (Middlebury College sponsored two additional students to work in conjunction with the nine NSF-REU funded students). The program was designed to provide undergraduates with first-hand experience conducting Psychological research, and to do so in an intensive environment during the summer. In my role as co-PI, I mentored five of the students as research interns in my laboratory, and co-coordinated all aspects of the summer program. More details regarding the nature of my mentoring philosophy for research students can be found in the accompanying “RUI Impact Statement.” Coordination of the REU program included arranging for students’ summer housing, arranging for student salary payments, arranging for off-campus students to be able to access campus buildings as well as the campus computer network, organizing a series of faculty talks given by REU faculty, and co-teaching a series of weekly seminars for the REU students. Topics for the weekly seminars ranged from data analysis with SPSS to careers in Psychology to designing and presenting research posters. Following the end of our summer REU program, I have continued to administer several aspects of our REU grant. First, I have kept track of the conference proposals that have been submitted by REU faculty to present the research with REU student co-authors. Second, I have encouraged REU faculty to invite their students to attend the conferences to which REU-based research presentations have been submitted, and have tracked the students who plan to attend conferences with REU faculty. This is an important duty for myself as one of the PIs on this grant, because our REU grant provides funding to help defray the cost of conference attendance.

Program Outcomes

The purpose of our REU program was to provide undergraduates with an opportunity to gain first-hand experience conducting research in collaboration with faculty, as well as to work collaboratively with other undergraduates in a laboratory environment. Among the main goals of our REU program, like all REU programs is to inspire promising undergraduates to pursue research at their home institutions and ultimately to attend graduate school. In evaluating our success in meeting these goals, our program experienced great success. Five of the nine summer students were recruited from institutions other than Middlebury, while four students came from Middlebury. Of these students, eight are currently conducting some form of independent research. All four Middlebury students are presently conducting senior theses, while four of the five non-Middlebury students are continuing their involvement in research. While it is still too early to measure the impact of this program on the students’ later decisions to enter graduate school, eight of the nine students indicated at the end of the program that the REU program had either inspired them to attend graduate school or solidified their desire to pursue graduate study.

At the conclusion of our REU program, we asked the students to fill out a survey evaluating all aspects of their research experience. Students’ evaluations were uniformly positive. Students repeatedly stated that working closely with a faculty member on actual research (as opposed to classroom demonstrations) was the most important part of their experience. Many students referred to the value of the program in clarifying their plans for further education, graduate training, and careers in psychology. On the items intended to evaluate overall satisfaction with the REU program, students generally rated the program highly. For example, when asked about the influence of the impact that the REU program had on preparing them to conduct future research, students rated the program highly (mean rating of 4.6, mode = 5 on a 5 point scale with 5 = “Significant impact” and 1 = “no impact”). Further, all nine of the students indicated that they would recommend our REU program to another student. Thus, we believe our REU program helped transfer the excitement faculty have for the research process to the students, and inspired them to continue their pursuit of research.

Finally, the REU program has already begun to be successful in terms of exposing the students to the dissemination process. All of the 2005 summer REU students are co-authors with REU faculty on

conference presentation proposals. While not all faculty have received indications of acceptance of those proposals, I feel that it is important that there has been such rapid turn-around on this first important dissemination step, which will allow the REU students to experience what it is like to present one's results at a professional conference. As of the submission of these conference proposals, all REU students expressed a desire to attend at least one conference and participate in the dissemination of their research. I view this as among the greatest successes of the program.

Project Description for RUI: Memory for perceiving information that was never experienced

The research proposed in this grant application seeks to understand the theoretical basis for the recollection of perceptual details in the absence of perceptual experience. Specifically, the research proposed in this grant application will evaluate one central research question that can explain the conviction people express in memories for events that never occurred: **are the representations underlying false memories associated with perceptual features that were present during encoding of related information, despite the fact that those memories were never perceived?** I propose to study this question via the use of a basic memory paradigm where subjects are presented with a series of items (e.g., *saddle, pony, mare, gallop, colt, stallion, rodeo, stable, trot, and unicorn*; referred to as *themes* below) that are all associatively related to a single word that is never presented during encoding (e.g., *horse*), referred to as the *lure item* (Roediger & McDermott, 1995). Extensive research in this paradigm has produced **five striking regularities**. First, subjects recognize and recall lures to a level that often approaches their level of recognition and recall for items they actually studied (Roediger & McDermott, 1995). Second, when participants are asked to provide judgments of recognition memory phenomenology (Tulving, 1985), they often indicate that they *remember* experiencing the presentation of lure items on the study list. Thus, rather than expressing a non-specific sense of familiarity with lure items, they maintain that they are able to recollect specific details of a lure's presentation on the study list, despite the fact that the lure was never perceived (e.g., Arndt & Reder, 2003; Roediger & McDermott, 1995). Third, lure errors seem to be impossible to eliminate, such that no known manipulation is able to eliminate them. For example, one study went to the lengths of giving subjects a strong warning about making errors, provided them with an example of a studied list and its associated lure prior to encoding, and offered participants the possibility to attribute items to sources other than the study list (e.g., that an item was related to study items, but was not shown on the list of words that the person studied). Yet, this study still found that subjects made errors to lure items well above the baseline rate of errors to unrelated foils (Multhaup & Connor, 2002). Fourth, subjects seem exceedingly willing to claim that a lure item occurred in one of two sources (e.g., a male or female voice) used to present its associates (Roediger, McDermott, Pisoni, & Gallo, 2004), suggesting that they truly believe the lure was presented to them. Finally, when subjects are asked to monitor their memories for the source from which a memory came (a task that typically enhances subjects' ability to distinguish between suggested and actually-experienced events; Lindsay & Johnson, 1989), they make more errors in recognition memory (Hicks & Marsh, 2001), and do not make fewer errors in recall (Hicks & Marsh, 1999). **In sum, these errors have proven extremely difficult to eradicate, and subjects seem to believe with strong conviction that they were experienced during a study list, to the point they will attribute the lure to a specific perceptual source.**

In order to address the question of whether or not the representations underlying false memories are associated with perceptual features present at encoding, I will use two different memory tasks that are sensitive to the overlap between perceptual features encoded at study and those present at retrieval: recognition memory tasks and perceptual implicit memory tasks. Therefore, the **first hypothesis tested is that lure items will be associated with perceptual encoding context, which will be revealed on both recognition memory tasks and perceptual implicit memory tasks**. In addition, the proposed research seeks to assess the extent to which associations between lure item representations and perceptual features encountered at encoding are strengthened by particular encoding circumstances. Prominent theories of memory errors suggest that the development of associations between lure items and perceptual features present during encoding is critically dependent on the extent to which encoding circumstances activate the

memory representations of lure items. Thus, despite the fact that lure items are by definition never perceived, it is possible that their representations in memory still become highly active when a number of associatively-related items are perceived during encoding. This high level of activation in turn may inspire the formation of associations between lure item representations and features of the encoding context, such as visual or auditory perceptual records. Therefore, the **second hypothesis tested is that when variables are manipulated that increase the activation of lure representations, the association between a lure's representation and perceptual encoding context will be stronger.**

To the extent that the results of these studies support the hypothesis that lure representations are associated with perceptual features of encoding context, it will provide an explanation for the five regularities described in the prior paragraph. Specifically, if lure representations are associated with context that occurs during encoding, no amount of warning or informing subjects about the structure of study lists in this paradigm will enable them to distinguish between studied and lure items perfectly, since encoding context is the primary dimension on which studied and unstudied items differ. Similarly, if lure representations are associated with encoding context, subjects' claims that a lure was presented by a specific source during encoding is an expected outcome, because subjects are retrieving authentic contextual information from memory in making source judgments about lure items, even though that information is misleading regarding the presentation source of the lure item.

In addition to the general goal of studying whether or not lures' representations are associated with perceptual encoding context, the studies outlined below seek to understand the conditions that influence the creation of associations between encoding context and lure representations. In particular, the studies below test predictions derived from activation-monitoring theory (Roediger, Watson, McDermott, & Gallo, 2001), one of the prominent explanations of false memory errors. Activation-monitoring theory suggests that two basic factors play a role in producing false memory. Activation-based processes lead to alterations in the level of activation that accrues at a lure's representation in memory, such that increased activation of a lure's representation produces increases in false memory. Activation-based processes tend to be influenced by factors that are manipulated at encoding, such as the number of associates of a lure item that are studied, the strength of the association between studied items and the lure, and the extent to which encoding emphasizes attention to the similarities among study items related to a lure (Roediger, et al., 2001). Monitoring-based processes are employed in an effort to verify the authenticity of a memory and tend to be utilized at retrieval (Roediger, et al., 2001). When subjects employ monitoring processes, they seek to retrieve evidence that establishes the studied status of a test item, such as a perceptual record of encoding the item. When they are able to retrieve such evidence, they will endorse the test item as studied. On the other hand, when they are unable to retrieve evidence that an item was encountered during encoding, they will tend to reject the test item as not studied.

Important for the research outlined below, activation-monitoring theory provides a testable claim regarding the circumstances under which lure representations can become associated with features that subjects encounter at encoding. The theory suggests that **to the extent that lure representations are activated during encoding, they will develop associations with features that are encountered at encoding.** Thus, it is possible for lure items to be associated with visual, auditory, and general contextual features encountered at encoding even though they are never actually presented to subjects. Further, **activation-monitoring theory suggests that those associations will be stronger when encoding circumstances activate a lure's representation strongly compared to when a lure's representation is activated weakly.** In the studies proposed below, both of these predictions will be systematically tested.

Prior Literature Suggesting Associations Occur Between Lures and Encoded Perceptual Features

Source Memory Tasks

Several recent results from the literature suggest that associations between unstudied lures and perceptual features of encoding context occur regularly, and that the basis for the strength of association is the amount of activation a lure's representation accrues during encoding. Two early studies suggested that

when subjects are asked to describe the characteristics of their memories, fewer perceptual details were provided when a lure item was endorsed as studied (Norman & Schacter, 1997; Mather, Henkel, & Johnson, 1997). However, these papers also reported that subjects *do* report some perceptual details when describing their basis for endorsing lure items as studied. Building on these results, Hicks and Hancock (2002) demonstrated that backward-associative strength was causally linked to lure item source memory judgments. Specifically, Hicks and Hancock presented lists of associatively related items in two different sources. One source (e.g., a male voice) was used to present the associates that were strongly related to that theme's lure, while a second source (e.g., a female voice) was used to present the associates that were weakly related to that theme's lure. Thus, half of each lure's related study items were presented in one of the two sources that subjects encountered during encoding. The results of two studies demonstrated that when a lure was erroneously endorsed as studied, subjects more often attributed the lure to the source used to present its stronger associates. These results therefore suggest that the attributions subjects make for lure items are based upon the extent to which their related study items activate the lure's representation, because stronger associates are more closely related to lure representations than weaker associates. In a second series of studies, Hicks and Starns (2006) showed that the influence of associative strength on lure source attributions maintained across various methods of presenting a lure's study items (blocked by source, blocked by category, but with sources intermixed, blocks and sources randomly intermixed).

These results are consistent with the idea that associations exist between encoding context (e.g., the voice in which study items are presented) and the representations of unstudied lures. In particular, the most straightforward interpretation of subjects' source attributions to lure items is that they are retrieving evidence from their memory that indicates a lure item was perceived. One mechanism that could give rise to such an attribution is that lure representations actually become associated with lure item representations as a consequence of having studied the lure's associates. While the source memory data can be interpreted this way, it is important to note that source memory data are necessarily reliant on subjects' introspective judgments that may or may not be indicative of the nature of the representations on which those attributions are based. The present proposal seeks to provide converging evidence for the conclusion that lure representations become associated with perceptual features of encoding context. The studies proposed here make use of objective means of assessing this hypothesis by testing memory with two tasks that are sensitive to manipulations that influence perceptual memory representations: recognition memory tasks and perceptual implicit memory tasks.

Perceptual Match Effects in Recognition Memory

The first way the hypothesis that lure representations develop associations with perceptual characteristics of study items will be tested is to examine perceptual match effects in recognition memory (Park, Arndt, & Reder, in press; Reder, Donavos & Erickson, 2002). Reder, et al. (2002) demonstrated that when study items were presented in an unusual perceptual format (e.g., gadzoox font), participants later showed facilitated recognition of those items when they were re-presented in the same perceptual format compared to items tested in a font that was used to present a different study item. Further, Reder, et al. (2002) demonstrated that this perceptual match effect was more pronounced when a font was used to present a single study item compared to multiple study items, a result that Reder, et al. (2002) termed the *font fan effect*. Reder, et al. (2002) suggested that this effect arose due to retrieval of perceptual information associated with an encoded memory representation, because the difference in recognition between items shown in a font used to present many other items and study items shown in a font that was used to present a single item only occurred when the studied font was reinstated at test. Similarly, Park, et al. (in press) presented study items in two perceptual formats, one visual (a font) and one auditory (a voice). Further, for each of the two features used to present study items, one was used to present many (twenty-four) study items, and one was used to present fewer (three) study items. For example, a study item might be presented in a voice that was used to present 24 total study items and a font that was used to present three total study items. On a later recognition memory test, Park, et al. (in press) reinstated one of the perceptual features shown with each study item, either the feature used to present many other items, or

the feature used to present few other study items. Park et al. found that recognition was enhanced when a test item was tested in a format used to present fewer other study items, consistent with Reder et al.'s (2002) finding of a font fan effect, and their attribution of that effect to retrieval-based processes.

In the series of studies proposed below, a similar approach is used to study the extent to which lure items develop associations with the perceptual features of their studied associates. Specifically, if lure representations develop associations with the perceptual features of related items that are studied, it should be possible to cue that association at retrieval by testing a lure in the perceptual format used to present its associates. In the studies proposed here, we utilize unusual-looking fonts to present a lure's associates and to present test items at retrieval. The use of unusual-looking fonts has two primary benefits. First, it allows each lure's associates to be presented in a different visual format, which in turn allows for a unique association to develop between a given lure's representation and the visual properties of its studied associates. This is important because Reder, et al. (2002) showed that font-matching effects in recognition memory rapidly diminish as a font is associated with an increasing number of concepts. Therefore, it is important to keep the number of concepts (themes) associated with each font to a minimum in order to arrive at a reasonable test of the existence of associations between lure representations and visual formats encountered at study. Second, demonstrating evidence that a lure's representation develops associations with an unusual-looking visual format encountered during encoding will provide strong evidence for the hypothesis that lure representations become associated with the perceptual features used to present their associates during encoding. Thus, while evidence exists that participants will believe that they heard a lure in a particular voice (e.g., Roediger, et al., 2004), it seems intuitively less likely that a participant will be able to imagine that a lure item was shown to them in a particular unusual-looking font with which they have limited or no experience (e.g., *wedgie* or *caveman*) simply because the lure's representation is highly active during encoding. Thus, even if study conditions lead to the lure coming to mind consciously during encoding, it is unlikely that the mental record of that occurrence will be confused with a mental record of an item that was actually viewed in an unusual-looking font. In comparison, when auditory sources are used to present study items during encoding, it is plausible that if participants consciously bring the lure to mind, the record of that occurrence will bear substantial similarity to the process of actually hearing study items being presented, because the voices used in such studies are neither unusual from the person's point of view nor would it be unlikely that the person would be able to imagine what the lure sounded like in one of the voices used to present study items.

Recent research conducted in my lab (Arndt, 2006) provides preliminary evidence 1) that visual features encountered at study are associated with lure representations 2) that the association between visual features of study items and lure items is influenced by the level of activation of lure representations during encoding, and 3) that the effects of these associations can be found on tests other than source memory tasks. In one study, lure items were either presented in a font used to present their studied associates (the *match* condition) or a font that was not encountered at encoding (the *mismatch* condition). This study demonstrated that lure item errors were greater in the match condition than in the mismatch condition, similar to what is found when study items are either tested in the same font used to present them during encoding or in a font not used to present them during encoding (Arndt, 2006; Reder, et al., 2002). In a second study, Arndt (2006) presented the study items related to each lure in two different fonts: one font was used to present the strong associates of the lure and a second font was used to present the weaker associates. On the memory test, each lure item was either presented in the font used to present its strong associates at encoding or the font used to present its weak associates at encoding. The results of this study suggested that presenting a lure item in the font used to present its strong associates during encoding produced more lure item errors than presenting a lure item in the font used to present its weak associates during encoding. Thus, **both studies outlined above suggest that false memory is increased when 1) unstudied lures are presented in a visual format used to present their related associates during encoding and 2) unstudied lures are presented in a visual format that was used to present their strongest associates during encoding.**

Perceptual Match Effects in Implicit Memory

The second way the hypothesis that lure representations develop associations with perceptual characteristics of study items will be tested is to utilize a perceptual implicit memory test. The implicit memory test chosen to examine perceptual format matching is the word fragment completion test, where test items are presented as words with letters deleted out of them (e.g., s_e_k_r for *speaker*). In this task, participants are asked to produce the first word that comes to mind that fits the fragment. The typical finding for a task such as word fragment completion, where the test cue is perceptual in nature, is that participants provide more correct solutions when the test cue corresponds to a word that was encountered at study, and when perceptual variables such as study modality (Roediger & Blaxton, 1987) and the font in which a study item was presented (Vaidya, Gabrieli, Verfaellie, Fleischman, & Askari, 1998) are the same on the implicit memory test as they were during the study episode. The first result, that participants provide more correct solutions when the test cue corresponds to a studied item than an unstudied item is referred to as *priming*. The latter two results, that priming is greater when study and test modality and perceptual format (e.g., font) match, have been taken as evidence that this implicit memory test involves the retrieval of perceptually-based memory representations (Roediger & McDermott, 1993). Enhanced implicit memory performance (*priming*) is therefore assessed by comparing the proportion of fragments corresponding to studied items that participants are able to complete to the proportion of fragments corresponding to unstudied items that participants are able to complete. A comparable measure of priming for lure items will be derived by comparing the proportion of lure items related to studied themes that are accurately completed to the proportion of lure items related to unstudied themes that are accurately completed.

Implicit memory tests have been used in several studies to probe the nature of the representations of lure items in general, and whether lure item representations contain perceptual details of their studied associates in particular (McKone & Murphy, 2000; Hicks & Starns, 2005). The application of implicit memory tests to study this question has taken two basic forms: tests that assess whether or not priming occurs for lure items (McDermott, 1997; McKone, 2004; Whittlesea, 2002), and tests that assess the extent to which priming for lure items is influenced by the match in modality from study to test (McKone & Murphy, 2000; Hicks and Starns, 2005). In an early study, McDermott (1997) found evidence of priming for lure items on the perceptual implicit memory tests of word stem completion and word fragment completion when the lure's associates were studied. Similarly, McKone & Murphy (2000) found evidence of priming for lure items using the perceptual implicit memory test of word stem completion, but only when study and test modality were the same. However, not all studies have shown evidence of priming for lure items. For example, McKone (2004) and Whittlesea (2002) failed to show evidence of priming for lure items related to studied themes in a lexical decision task. Finally, Hicks and Starns (2005) assessed the extent to which priming for lure items occurred when study and test modality matched and did not match. When they used a word stem completion test, Hicks and Starns (2005) found evidence of priming for lure items, and that the amount of priming did not differ across visual (study-test modality match) and auditory (study-test modality mismatch) study of a lure's associates. Thus, Hicks and Starns (2005) did not replicate the result of McKone and Murphy (2000) that priming for lure items was sensitive to the match between the modality in which a lure's associates were studied and the modality in which the lure was tested. Further, Hicks and Starns (2005) examined the influence of study-test match for lure items using a second implicit memory test, perceptual identification, where subjects are asked to identify rapidly-presented test items (e.g., for 33 msec). Using this test, Hicks and Starns (2005) failed to find evidence of priming for lure items, regardless of whether their associates were studied in the auditory or the visual modality.

Thus, the evidence from implicit memory tests poses a critical challenge to the hypothesis that lure representations develop associations with perceptual features that are used to present related theme items during encoding. Specifically, the hypothesis that lure representations are associated with perceptual records of the experience of their studied associates would expect that lures would show priming on perceptual implicit memory tests, and that this priming would be influenced by the match between the

perceptual modality in which a lure's associates were studied and the modality in which the lure was tested. However, both of these effects have been inconsistent in the existing literature.

The studies proposed below will provide another test of the hypothesis that lure representations are associated with the perceptual encoding context of their studied associates by examining font-match effects (e.g., Vaidya, et al., 1998). Font-match effects have been cited as providing important evidence for perceptual forms of implicit memory. In addition, the use of font-match effects offers the potential to present each lure's associates at study in a different perceptual format. This procedural dimension is important because it is possible that prior studies have shown inconsistent results because of the accrual of extensive perceptual interference. Specifically, when the same perceptual format (modality) is used to present a large number of study items, the ability of that perceptual format to act as a memory retrieval cue tends to become weakened (Watkins & Watkins, 1975). Thus, it is possible that prior studies using modality as a perceptual cue have failed to find evidence of a modality match effect for lure items because of extensive levels of perceptual interference caused by the use of only two perceptual formats.

The general method that will be used to accomplish the study of font match effects for lure items will be to present study items for each theme in a single font, and to have fonts differ across themes in order to limit the accrual of perceptual interference. Specifically, the proposed studies make use of unusual-looking fonts that are unlikely to have been experienced by subjects previously (e.g., *caveman* and *wedgie*) to present study items and test items on the implicit memory test. The use of these fonts limits the accrual of perceptual interference in two fundamental ways. First, the presentation of each theme in a different visual format limits the number of concepts with which each perceptual format is associated. Second, subjects are likely to have limited prior exposure to the fonts that will be used to present study and test items, which makes it more likely that each font will have little to no pre-experimental exposure, and therefore little extra-experimental perceptual interference. On a later perceptual implicit memory test, study items and lure items will be presented in either a match condition or a mismatch condition. In the match condition, study items will be shown in the font used to present them at study, while in the mismatch condition, study items will be shown in a font used to present a different theme in the study list. Similarly, lure items in the match condition will be shown in the same font that was used to present their associates at study, while lure items in the mismatch condition will be shown in a font that was presented at study, but was not used to present their associates. Therefore, if lure items show greater levels of priming in the match condition compared to the mismatch condition, that result would provide evidence that lure representations were associated with the perceptual features of its studied associates.

Although some authors suggest that the test used in the studies proposed below (word fragment completion) is susceptible to influences from processes other than perceptual fluency (e.g., Hicks & Starns, 2005), there are two reasons why this test is the most appropriate for the studies proposed here. First, a different unusual-looking font will be used to present the study items from each theme in order to reduce the build up of perceptual interference within the study list. In order to present each theme in a different unusual-looking font, it will be necessary to select a relatively large number of fonts for use, which will inevitably require substantial variability in the visual properties of fonts, which could lead to differences in how participants perceive each font, in effect increasing performance variability. Second, based on my experience selecting fonts to use in the studies described in the preliminary studies section (Arndt, 2006; Arndt & Reder, 2003), some of the unusual-looking fonts that will be used in these studies will be sufficiently challenging to read such that participants will be less able to perform some perceptual implicit memory tasks, such as perceptual identification or lexical decision. Thus, in order for subjects to be able to perform an implicit memory task where test items are presented in a perceptual format that minimizes perceptual interference, it will be necessary for participants to have somewhat more time to perceive each test item than is afforded by a perceptual identification task or a lexical decision task, where the ability to rapidly perceive stimuli in a perceptually degraded form is an essential component of performance.

Finally, in all of the perceptual implicit memory studies proposed below, several measures will be taken to minimize the potential influence of explicit memory contamination on implicit memory performance (Schacter, Bowers, & Booker, 1989). First, a 10 minute retention interval will be inserted

between the study list and the implicit memory test. This will enable me to present the study and test tasks as different tasks in a series of tasks that the subject is asked to complete, thereby reducing the chance that the subject will realize the connection between the study and test tasks, and the consequent chance that the subject will attempt to use intentional retrieval. Second, the majority of the test items (75%) will be items that subjects did not encounter during study, again reducing the potential to notice the connection between the study list and the implicit memory test. Third, following the completion of the study, subjects will be given a questionnaire to assess their awareness of a connection between the study task and the implicit memory test, as well as their use of intentional retrieval to complete items on the implicit memory test (e.g., McKone & Murphy, 2000; Mulligan, 2002). This questionnaire will be used to ensure that subjects did not use explicit memory retrieval to complete the word fragments on the implicit memory test, thereby turning it in to an explicit cued recall task. Thus, the results of the implicit memory studies will be analyzed using 1) all subjects who participate in a given implicit memory study, and 2) excluding subjects who report using intentional retrieval in order to complete portions of the implicit memory task. The latter analyses will provide the optimal assessment of perceptual implicit memory for lure items, because participants who do not use explicit memory retrieval are likely to be using implicit memory alone.

Proposed Studies

The studies proposed below will further examine the general notion that features of encoding context become associated with lure representations. These studies use two basic types of memory tests that have been shown to be sensitive to perceptual overlap between encoding and retrieval: recognition memory tasks and perceptual implicit memory tasks. One variable that will be manipulated in all of the studies proposed below is whether or not lure representations are presented in a font that matches the presentation characteristics of their studied associates. This variable allows for a test of the **first hypothesis** these studies test: **lure representations become associated with perceptual features of encoding context via study of their related associates.** Specifically, if it is the case that lure representations are associated with perceptual features that were present at encoding, one would expect lure errors and lure priming to be greater in when lures are presented in the same font that was used to present their studied associates.

A second class of variables will be manipulated in these studies, with the goal of examining the role that lure activation plays in the production of associations between lure representations and perceptual features present during encoding. Two of the variables that will be manipulated in order to influence activation processes are the product of inherent characteristics of study items, while the third variable that will be used to influence lure activation results from strategic encoding processes. The variables that are the product of inherent study item characteristics are the number of associates studied (Arndt & Hirshman, 1998; Gallo & Roediger, 2003; Mintzer & Griffiths, 2001) and the strength of the association between the study items and lures (i.e., associative strength; Hicks and Hancock, 2002; McEvoy, Nelson, & Komatsu, 1999; Roediger, et al., 2001). The variable that is a product of strategic encoding processes involves a manipulation where some themes will be studied by asking subjects to attend to the similarities among the items within a theme, while other themes will be studied by asking subjects to generate a property that distinguishes each study item from all others in the theme (Hunt & Elliot, 1981; McCabe, Presimanes, Robertson, & Smith, 2004; Roediger, et al., 2001). The logic behind using this study task is that lure activation should be increased to the extent that subjects engage in processing of the similarities among study items (McCabe, et al., 2004). Thus, by manipulating subjects' encoding strategies to encourage or discourage relational processing, it is possible to influence the activation of a lure's representation while keeping item characteristics constant across conditions. The manipulation of variables that influence activation processes allows for a test of the **second hypothesis** these studies test: **when variables are manipulated that increase the activation of lure representations, the association between a lure's representation and perceptual encoding context will be stronger.** If it is the case that this hypothesis is viable, the magnitude of the match effect for lure errors and lure priming should be greater in the condition where the lure was activated more at encoding (i.e., when more associates were studied, when a

perceptual source is associated with the strong associates of a lure, and when a theme is studied in a way that emphasizes the relationships among its study items.).

To the extent that evidence is found supporting the hypothesis that lure representations are associated with presentation context, these studies will suggest a reason for the occurrence of the five striking regularities noted in the introduction. Recall that these regularities all provide evidence that participants behave in ways that indicate they believe their memories for lure items are authentic, leading them 1) to endorse lures as studied with a high probability; 2) to believe that they recollect the occurrence of the lure on the study list; 3) to be unable to eliminate lure errors even when they are warned against committing errors and given extensive information about the nature of the paradigm; 4) to attribute a lure's presentation to a source that was used to present study items; and 5) to commit more errors to lure items when asked to make source judgments rather than old-new recognition judgments. All of these major results make considerable sense if it is the case that lure items develop associations with information present during encoding, because all of the above results are consistent with the notion that participants experience lure items as similar to items that are actually studied. According to the hypothesis that lures develop associations with features of encoding context, these behaviors simply derive from the fact that lures *are* represented in memory in a way that is strikingly similar to other memories that were actually perceived, so it is not surprising that people behave as if lure items *were* perceptually experienced.

In general, if the hypothesis that lure representations develop associations with perceptual features that are used to present their studied associates is viable, evidence of those associations should be revealed on any memory test that is sensitive to such an association. Thus, when given a recognition memory test, participants' judgments that a lure was presented should be increased when the lure is presented in a perceptual format that was used to encode a lure's associates. Similarly, when given an implicit memory test that is sensitive to perceptual matching from encoding to retrieval, participants should show more priming when a lure is presented in a visual format used to present its associates relative to when it is presented in a visual format that does not match the format used to present its associates. Finally, both of these effects should be enhanced when encoding circumstances produce greater levels of activation of a lure's representation, since lure activation is purported to play an important role in determining the strength of the association between a lure's representation and the perceptual features used to present a lure's associates.

Perceptual Match Effects in Recognition Memory

As described above, the influence of associative strength on perceptual match effects in recognition memory was recently studied by Arndt (2006). Arndt (2006) found that participants made more lure false alarms when lure items were presented in the font used to present the lure's strong associates relative to when lure items were presented in the font used to present the lure's weak associates.

Sample Size: As outlined in the *Data Analysis* section below, the sample size used in studies of perceptual match effects in recognition memory will be $N = 64$. This sample size is expected to provide statistical power in excess of .90 given the size of study-test match effects found in prior studies (Arndt, 2006).

Proposed Study 1: Manipulating the number of associates studied to increase lure activation

Proposed Study 1 will further examine the extent to which lure representations are associated with perceptual information encoded with their studied associates by manipulating the number of associates of a lure that are studied, as well as whether the font presented at retrieval matches the font used to present the lure's associates.

Materials

Stimulus items will be selected from the Nelson, McEvoy, and Schreiber (1998) database. Stimuli will be selected such that all items within a theme produce the same lure item with a nonzero probability. Further, themes will be selected such that study items are those that have the highest probability of

producing the lure item, while ensuring that there is no overlap across themes (i.e., that no study items are repeated and that no lure item appears in the study items for a different theme).

Study and test stimuli will be presented in a variety of unusual-looking fonts. My lab currently has selected 144 fonts from various internet sources for use in font manipulation studies. The fonts that will be utilized in this experiment are ones that participants are unlikely to have encountered previously, do not resemble fonts one typically sees in everyday life (e.g., Helvetica, Courier), are easily readable, and are similar in size to one another. The subset of these fonts that are most easily distinguished from one another will be used to present study and test items.

Design and Procedure

A standard recognition memory procedure will be utilized in this experiment. The two experimental factors in this study will be implemented by manipulating the characteristics of the study and test lists. In the study list, all studied themes will be presented for participants to encode, with some themes having eight associates presented and some themes having two associates presented. On the recognition memory test, participants will be instructed that their task is to discriminate between studied and unstudied items. In the test list, the match vs. mismatch condition will be manipulated by presenting test items either in a font used to present that theme at study (the *match condition*) or a font used to present a different theme's study items at study (the *mismatch condition*). Thus, lure items assigned to the match condition will be presented in the font that was used to present its studied associates, and lure items assigned to the mismatch condition will be presented in a font that was shown during the study list, but was not used to present that lure's associates.

Proposed Study 2: Manipulating encoding strategies to increase lure activation

Proposed Study 2 will further examine the extent to which lure representations are associated with perceptual information encoded with their studied associates by manipulating the encoding strategies used by participants during study, as well as whether the font presented at retrieval matches the font used to present the lure's associates.

Method, Design, and Procedure

The method, design, and procedure will be similar to that of the eight associate condition in Proposed Study 1 described above, with the exception that Study Task (attend to similarities vs. attend to differences; McCabe, et al., 2004) will be manipulated instead of the number of associates studied. Study Task will be manipulated by presenting all of the study items in a theme simultaneously on the computer screen and then either asking participants to generate one property that all eight study items have in common (similarity study task) or one property that differentiates each study item from the other seven (difference study task). Subjects will be allotted 40 seconds per theme to perform the similarity or difference study task. Further, study task will be blocked within each study list, such that subjects will perform the similarity study task on half of the themes and the difference study task on the other half of the themes. All other procedural details from the eight associate condition of Proposed Study 1 will be maintained in this study (i.e., how the match and mismatch conditions are constructed for study items and lure items).

Perceptual Match Effects in Implicit Memory

Sample Size: As outlined in the *Data Analysis* section below, the sample size used in studies of perceptual match effects in recognition memory will be $N = 128$. This sample size is expected to provide statistical power in excess of .80 assuming a medium effect size (Cohen, 1988).

Proposed Study 3: Manipulating Strong vs. weak associates to increase lure activation

Proposed Study 3 will examine the extent to which lure representations are associated with perceptual information encoded with their studied associates by manipulating the strength of the associates that are

studied with a particular visual format. This study will therefore explore the extent to which recognition memory (Arndt, 2006) and perceptual implicit memory show similar effects of presenting a lure item in a font that was used to present that lure's strong or weak associates during encoding.

Materials

Stimulus items will again be selected from the Nelson, McEvoy, and Schreiber (1998) database. For each theme, four strong associates and four weak associates will be selected. Strong associates will be defined as the four associates that produce the lure with the greatest probability in free association, with a minimum probability of producing the lure item in free association of .15. Weak associates will be defined as four associates that produce the lure with a probability between .02 and .10 in free association.

Design and Procedure

A standard implicit memory procedure will be utilized in this study. For each studied theme, all eight study items in that theme will be presented (four strong associates and four weak associates). The strong associates of each lure will all be presented in a single font, while the weak associates of each lure will all be presented in a second font. Strong and weak associates will be presented in a blocked manner, and it will be ensured that an equal number of themes have their weak and strong associates presented first. Different fonts will be used across themes to minimize perceptual interference. Following the presentation of the study list, participants will be engaged in distractor activity for 10 minutes. The distractor activity will be unrelated to the study task, and will be designed to encourage subjects to believe that they are completing a series of mental tasks that are unrelated to one another.

Implicit memory test items will be constructed by deleting half of the letters in each test word. Participants will be instructed to attempt to solve the word fragment by providing the first word that comes to mind that completes the fragment. No mention will be made of the connection between the fragment completion task and the study task. Test lists will consist of one studied item per theme, the lure item for each studied theme, one unstudied associate for each theme not presented at study, and the lures corresponding to the themes not presented at study. Therefore, 75% of the items occurring on the implicit memory test will be unstudied, which should help serve to reduce the likelihood that subjects realize the connection between the study list and the implicit memory test. Half of the studied items presented on the test list will be strong associates and half of the studied items presented on the test list will be weak associates of the lure items. Further, half of the studied items presented on the test list will be presented in the font that they were presented in at encoding (the *match condition*), while the other half of the studied items presented on the test list will be presented in a font that was used to present a different theme's study items at encoding (the *mismatch condition*). Similarly, half of the lure items will be presented in a font used to present their associates at encoding (the *match condition*), while the other half of the lure items will be presented in a font used to present the associates of a different lure at encoding (the *mismatch condition*). Further, within the match and mismatch conditions, half of the lure items will be presented in a font used to present strong associates at study and half of the lure items will be presented in a font used to present weak associates at study. Thus, lures in the match condition will be presented in either the font utilized to present their strong associates or their weak associates at study. Unstudied theme items and lure items related to unstudied themes will be presented in order to derive an estimate of the baseline level of completion of word fragments for theme items and for lure items. Half of the unstudied theme items and half of the lure items related to unstudied themes will be presented in a font used to present strong associates during study, and the other half of each item type will be presented in a font used to display weak associates during study.

Following the completion of the test list, participants will be given a questionnaire to assess their awareness of the connection between the study task and the implicit memory test modeled after Mulligan (2002). This awareness questionnaire will be used to divide subjects into groups that report awareness of the connection between the study task and the implicit memory task. In particular, it will be used to assess the extent to which subjects who were unaware of the connection between the study task and the implicit memory task show evidence of font-matching effects for lure items.

Proposed Study 4: Manipulating the number of associates studied to increase lure activation

Proposed Study 4 will further examine the extent to which lure representations are associated with perceptual information encoded with their studied associates by manipulating the number of associates of a lure that are studied, as well as whether the font presented at retrieval matches the font used to present the lure's associates. This study, in conjunction with Proposed Study 1, will explore the extent to which recognition memory and perceptual implicit memory show similar effects of these two manipulations.

Method, Design, and Procedure

The method, design, and procedure will borrow elements of the methodology of Studies 1 and 3 outlined above. The manipulations used in this study will be the same as those used in Study 1 (manipulations of number of associates studied and study-test match). Similarly, the distraction task following the study task, the implicit memory task, and the post-test awareness assessment used in this study will be constructed using the same parameters as those used in Study 3.

Proposed Study 5: Manipulating encoding strategies to increase lure activation

Proposed Study 5 will further examine the extent to which lure representations are associated with perceptual information encoded with their studied associates by manipulating encoding strategies to influence lure activation, as well as whether the font presented at retrieval matches the font used to present the lure's associates. This study, in conjunction with Proposed Study 2, will explore the extent to which recognition memory and perceptual implicit memory show similar effects of these two manipulations.

Method, Design, and Procedure

The method, design, and procedure will be similar to that of Studies 3 and 4 described above, with the exception that study task (attend to similarities vs. attend to differences) will be manipulated instead of number of associates studied or associative strength. Study Task will be manipulated in the same way as it was in Proposed Study 2, where study task was blocked, each theme was presented in a single font, and a different font will be used for each theme. All other procedural details from Proposed Studies 3 and 4, which also will use implicit memory tasks, will be maintained in this study.

Data Analysis

Data analysis for these studies will be accomplished via within-subjects factorial Analysis of Variance (ANOVA). All of the designs outlined below are 2 x 2 within-subjects designs. As noted above, evidence supporting the contention that lure items are associated with perceptual features encountered at encoding will come in the form of an interaction between a factor that affects lure representation activation (number of associates studies, associative strength of study items to lures or the nature of the study task subjects engage in) and study-test matching of fonts.

The sample sizes for the proposed studies were based on the size of font match effects found in prior studies conducted in my laboratory. Briefly, the effect size measure f (Cohen, 1988) was used to derive an estimate of the size of the effect of the font match effect for lure items from Experiment 2 of Arndt (2006) and the size of the effect of the strong vs. weak associate font match effect for lure items from Experiment 4 of Arndt (2006). The observed effect size for the effect of font match was $f = .41$ and the observed effect size for the effect of strong vs. weak associate font match was $f = .61$. In order to plan samples sizes as conservatively as possible, I chose to use the smaller of the two effect sizes as the benchmark for power calculations. Given an effect size of $f = .41$, I then used G-POWER (Faul & Erdfelder, 1992) to estimate the optimal sample size for power to be above .90. The sample size that fit best with a full counterbalance for the recognition memory studies above was 64 subjects, which generates *a-priori* power to detect an effect size of $f = .41$ or greater in excess of .94. Given that implicit memory effects tend to be smaller and less reliable (Buchner & Wippich, 2000), as well as my intent to divide subjects performing the implicit memory task in to groups based on their awareness of the connection between the study list and implicit

memory test, I doubled the target sample size for the implicit memory studies to 128 subjects in an effort to ensure an adequate test of the hypotheses in the implicit memory studies. Given this sample size, the power to detect a medium-sized effect ($f=.25$; Cohen, 1988) is above .80. Thus the total sample size across the six studies proposed below is 512 (64 in each of two studies of recognition memory and 128 in each of three studies of implicit memory). In my experience, two lab students (the number requested per school year) can generally recruit and test approximately 80-100 subjects per semester between the Introduction to Psychology subject pool and recruitment via flyers advertising \$10 for participation in an experiment, given that their duties also include stimulus list construction and experiment programming. Further, experience the past two summers suggests that summer research students should be able to recruit and test between 40 and 50 subjects each summer. Thus, over the term of this grant proposal (3 academic years and three summers), it is plausible to collect the required number of subjects in order to complete these studies, given the expected pace of approximately 200 subjects per year including summer data collection. If data collection proceeds at a considerably more rapid pace than is estimated, the extra subjects will be tested in the implicit memory experiments to maximize the observed statistical power in those studies.

Challenges and Alternative Hypotheses

The biggest challenge to completing all of these studies during the grant period is the ability to recruit the large number of subjects needed to participate in these studies. Considering these studies will be conducted at a small school, there are a limited number of students available to participate in research each semester. Over the course of the school year, it is typically possible to recruit approximately 80-100 subjects per semester based on my experience since arriving at Middlebury. The typical number of students enrolled in Introduction to Psychology each term is approximately 100-120. Over my first 3.5 years at Middlebury, I have adjusted to the challenges of collecting data at an institution of this size, which allows me to maximize my lab's use of the subject pool during each semester. In order to maximize our data collection, we typically have a series of studies planned, programmed, and therefore ready to run at the start of each semester. This allows my research students and I to collect data the entire duration of the semester, without having to take time off between the completion of one study and the beginning of running the next study. Thus, I believe that there will be a sufficient number of participants to fulfill the sample sizes for the five studies proposed in this grant. In the event there are fewer introductory psychology students available for participation in a particular semester, part of the budget request for this grant proposal is for funding to pay research participants. Thus, collecting a sample size adequate to conduct these studies should be possible.

The design of these studies also allows for the testing of alternate hypotheses regarding the genesis of lure errors. In this basic design, the mismatch condition provides a sense for the differences in activation accrued by lure representations as a function of the variables that should influence lure activation, while the match condition provides a sense for the extent to which there is an association between lure representations and the perceptual features of its associates that is stronger when encoding conditions increase the activation of lure representations. Thus, the design of each of these studies also tests the hypothesis that the effects of the encoding factors manipulated (number of associates studied, associative strength and study strategies) only increase the activation of lures' representations, and do not lead to the formation of associations between the perceptual features of a lure's studied associates and the lure's representation. If manipulating a factor that increases lure activation does not lead to the formation of associations between lure representations and perceptual features encountered during the encoding of the lure's associates, then one would expect the effects of those encoding factors on lure errors in recognition memory and lure priming to be comparable across the match and mismatch conditions. In contrast, the hypothesis that lure representations develop associations with the perceptual format of their studied associates predicts that there will be an interaction between the factor that increases lure activation and study-test match for lure errors in recognition memory and lure priming.

Studying lure errors and lure priming using visual sources and font-matching at test also provides a method for arguing against the hypothesis that the only reason lures show evidence of being experienced during encoding is because they come to mind consciously while their associates are being studied. Specifically, if the basis for higher rates of errors to lure items is simply that lures come to mind during the encoding of associatively-related lists of items, there is no reason to believe that conscious thinking of the lure will lead it to become associated with visual features such as the unusual-looking fonts used in the present studies. Thus, this hypothesis predicts that whether or not the font used to present a lure at retrieval matches the font used to present its associates should not dramatically influence the level of errors to lure items that are committed or the amount of priming for lure items.

Finally, in concluding, it is important to note that if one type of task fails to show evidence consistent with associations between studied information and perceptual features encountered at encoding, that result will still possess theoretical implications, in addition to suggesting a direction for future research. Consider the possibility that these studies find that recognition memory tests consistently show evidence suggesting the existence of associations between lure representations and perceptual features of their related study items, but implicit memory tests do not show comparable evidence. This result would suggest that a profitable direction for future research would be to investigate the kinds of retrieval processes that differ between perceptual match effects in recognition memory and perceptual match effects in implicit memory retrieval. Thus, it would then become important to attempt to understand the critical differences between retrieval matching in recognition memory and implicit memory. For example, it may be that in recognition memory, subjects are inspired to utilize monitoring processes, and those monitoring processes make contact with evidence in memory that a lure was experienced in a particular perceptual format, but that implicit memory retrieval does not engage the types of retrieval processes that are necessary for contacting associations between perceptual features from the encoding environment and lure representations, leaving implicit memory performance unaffected by perceptual matching for lure items. Thus, regardless of the ultimate outcome of the line of research outlined in this proposal, there are additional questions that can be asked in order to explore further dimensions of the basis for why people so often believe strongly that illusory memories were experienced and can be recollected, often in vivid perceptual detail.

Broader Impacts

The broader impacts of conducting this research concern both the role that undergraduate research students will play in conducting the proposed projects, as well as the application of the results of these studies to broader societal issues. First, undergraduate researchers will play an integral role in conducting these studies, given that the PI is at a small liberal arts college that does not have graduate students. Thus, all interactions about these research projects will be directly between the PI and the undergraduate research students working on these studies. As outlined in the *RUI Impact Statement*, the PI engages undergraduates in research at a high level, integrally involving students who collaborate with him on research in all elements of the research process, from study conceptualization to dissemination of results, and all of the steps in between. Thus, the act of completing these studies will serve to further undergraduate education by exposing undergraduates to the research process at a deep level. In addition, the PI will make every effort to recruit qualified candidates from underrepresented groups. The group that he has experienced the most success in recruiting is female students, given that female students make up the plurality of undergraduate psychology majors at Middlebury College. Thus, conducting this research will at a minimum expose female undergraduates, a traditionally underrepresented group in the sciences, to research first hand. Further, the PI will continue to seek out highly motivated undergraduates from other groups that are traditionally underrepresented in the sciences by both inviting motivated students from his prior courses to collaborate on this research, as well as by seeking out recommendations from department colleagues for motivated students.

The broader impacts of this research also extend in to the realm of applied work in the areas of eyewitness memory and therapeutic practice within clinical psychology, where considerable work over the

past 30 years suggests that memory can be strikingly inaccurate. Two particularly relevant domains are the current debate over the authenticity of recovered memories of past traumatic events (e.g., childhood sexual abuse), and the authenticity of eyewitness memory reports. Recovered memories and eyewitness memory reports are often accompanied by vivid recollections, such that people reporting such memories claim to be able to see events in their minds' eye, and can describe those events by providing rich perceptual detail. Such vivid recollections are often taken as a marker of a memory's authenticity by laypersons on juries (Cutler, Penrod, & Thomas, 1988) as well as some clinicians (Loftus & Ketcham, 1994). However, studies have demonstrated that such rich perceptual recollections can be created from suggestive probing of a person's memory, suggesting vivid perceptual recollection may not be an infallible cue for distinguishing between memories that are and are not authentic (Loftus, 1993). The results of these studies will provide valuable information about the nature of the memory representations underlying recollection of details that were never directly experienced. Thus, these studies investigate one plausible explanation for why people believe they experienced events that never occurred, and often provide perceptually-rich narratives to accompany those accounts in eyewitness memory and recovered memory accounts – that their memory representations actually contain authentic perceptual information, even though the events in question were never directly experienced.

References Cited

- Arndt, J. (2006). Distinctive information and false recognition: The role of encoding and retrieval processes. *Journal of Memory and Language*, 54, 113-130.
- Arndt, J., & Hirshman, E. (1998). True and false recognition in MINERVA2: Explanations from a global-matching perspective. *Journal of Memory and Language*, 39, 371-391.
- Arndt, J., & Reder, L. M. (2003). The effect of distinctive visual information on false recognition. *Journal of Memory and Language*, 48, 1-15.
- Buchner, A., & Wippich, W. (2000). On the reliability of implicit and explicit memory measures. *Cognitive Psychology*, 40, 227-259.
- Cohen, J. (1988). *Statistical power analysis for the behavioral sciences* (2nd edition). Hillsdale, NJ: Lawrence Erlbaum Associates.
- Cutler, B.L., Penrod, S.D., & Stuve, T.E. (1988). Juror decision making in eyewitness identification cases. *Law & Human Behavior*, 12, 41-55.
- Faul, F., & Erdfelder, E. (1992). GPOWER: A priori, post-hoc, and compromise power analysis for MS-DOS. [Computer program]. Bonn, FRG: Bonn University, Department of Psychology.
- Gallo, D.A., & Roediger, H.L., III (2003). The effects of associations and aging on illusory recollection. *Memory & Cognition*, 31, 1036-1044.
- Hicks, J.L., & Hancock, T.W. (2002). Backward associative strength determines source attributions given to false memories. *Psychonomic Bulletin & Review*, 9, 807-815.
- Hicks, J.L., & Marsh, R.L. (1999). Attempts to reduce the incidence of false recall with source monitoring. *Journal of Experimental Psychology: Learning, Memory and Cognition*, 25, 1195-1209.
- Hicks, J. L., & Marsh, R. L. (2001). False recognition occurs more frequently during source identification than during old-new recognition. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 27, 375-383.
- Hicks, J.L., & Starns, J.J. (2005). False memories lack perceptual detail: Evidence from implicit word-stem completion and perceptual identification tests. *Journal of Memory and Language*, 52, 309-321.
- Hicks, J.L., & Starns, J.J. (2006). The roles of associative strength and source memorability in the contextualization of false memory. *Journal of Memory and Language*, 54, 39-53.
- Hunt, R. R., & Einstein, G. (1981). Relational and item-specific information in memory. *Journal of Verbal Learning and Verbal Behavior*, 20, 497-514.

- Lindsay, D.S., & Johnson, M.K. (1989). The eyewitness suggestibility effect and memory for source. *Memory & Cognition*, 17, 349-358.
- Loftus, E.F. (1993). The reality of repressed memories. *American Psychologist*, 48, 518-537.
- Loftus, E.F., & Ketcham, K. (1994). *The myth of repressed memory: False memories and allegations of sexual abuse*. St. Martin's Press.
- Mather, M., Henkel, L.A., & Johnson, M.K. (1997). Evaluating characteristics of false memories: Remember/know judgments and memory characteristics questionnaire compared. *Memory & Cognition*, 25, 826-837.
- McCabe, D.P., Presmanes, A.G., Robertson, C.L., & Smith, A.D. (2004). Item-specific processing reduces false memories. *Psychonomic Bulletin & Review*, 11, 1074-1079.
- McDermott, K.B. (1997). Priming on implicit memory tests can be achieved through presentation of associates. *Psychonomic Bulletin & Review*, 4, 582-586.
- McEvoy, C.L., Nelson, D.L., & Komatsu, T. (1999). What is the connection between true and false memories? The differential roles of interitem associations in recall and recognition. *Journal of Experimental Psychology: Learning, Memory and Cognition*, 25, 1177-1194.
- McKone, E. (2004). Distinguishing true from false memories via lexical decision as a perceptual implicit test. *Australian Journal of Psychology*, 56, 42-49.
- McKone, E., & Murphy, B. (2000). Implicit false memory: Effects of modality and multiple study presentations on long-lived semantic priming. *Journal of Memory and Language*, 43, 89-109.
- Mintzer, M.Z., & Griffiths, R.R. (2001). False Recognition in Triazolam-Induced Amnesia. *Journal of Memory and Language*, 44, 475-492.
- Mulligan, N.W. (2002). The effects of generation on conceptual implicit memory. *Journal of Memory and Language*, 47, 327-342.
- Multhaup, K. S., & Conner, C. A. (2002). The effects of considering non-list sources on the Deese-Roediger-McDermott memory illusion. *Journal of Memory and Language*, 47, 214-228.
- Nelson, D. L., McEvoy, C. L., & Schreiber, T. A. (1998). The University of South Florida word association, rhyme, and word fragment norms. <http://www.usf.edu/FreeAssociation/>.
- Norman, K. A., & Schacter, D. L. (1997). False recognition in younger and older adults: Exploring the characteristics of illusory memories. *Memory & Cognition*, 25, 838-848.
- Park, H., Arndt, J., & Reder, L. (in press). A contextual interference account of distinctiveness effects in recognition. *Memory & Cognition*.

- Reder, L. M., Donavos, D., & Erickson, M. A. (2002). Perceptual match effects in direct tests of memory: The role of contextual fan. *Memory & Cognition*, 30, 312-323.
- Roediger, H.L., III, & Blaxton, T.A. (1987). Effects of varying modality, surface features, and retention interval on priming in word-fragment completion. *Memory & Cognition*, 15, 379-388.
- Roediger, H. L., III., & McDermott, K. B. (1993). Implicit memory in normal human subjects. In F. Boeller & J. Grafman (Eds.) *Handbook of Neuropsychology*, Vol. 8, (pp. 63-131). Amsterdam: Elsevier.
- Roediger, H. L., III., & McDermott, K. B. (1995). Creating false memories: Remembering words not presented in lists. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 21, 803-814.
- Roediger, H.L., III., McDermott, K.B., Pisoni, D.P., & Gallo, D.A. (2004). Illusory recollection of voices. *Memory*, 12, 586-602.
- Roediger, H.L., III, Watson, J.M., McDermott, K.B., & Gallo, D.A. (2001). Factors that determine false recall: A multiple regression analysis. *Psychonomic Bulletin & Review*, 8, 385-407.
- Rotello, C.M., Macmillan, N.A., & Van Tassel, G. (2000). Recall to reject in recognition: Evidence from ROC curves. *Journal of Memory and Language*, 43, 67-88.
- Schacter, D.L., Bowers, J., & Booker, J. (1989). Intention, awareness and implicit memory: The retrieval intentionality criterion. In S. Lewandowsky, J.C. Dunn, & K. Kirsner (Eds.) *Implicit memory: Theoretical issues* (pp. 47-65). Hillsdale, NJ: Erlbaum.
- Tulving, E. (1985). Memory and consciousness. *Canadian Psychologist*, 26, 1-12.
- Vaidya, C.J., Gabrieli, J.D., Verfaellie, M., Fleischman, D., & Askari, N. (1998). Font-specific priming following global amnesia and occipital lobe damage. *Neuropsychology*, 12, 183-192.
- Watkins, O.C., & Watkins, M.J. (1975). Buildup of proactive inhibition as a cue-overload effect. *Journal of Experimental Psychology: Human Learning & Memory*, 1, 442-452.
- Whittlesea, B.W.A. (2002). False memory and the discrepancy-attribution hypothesis: The prototype-familiarity illusion. *Journal of Experimental Psychology: General*, 131, 96-115.

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Education

Post-doctoral Carnegie Mellon University. Cognitive Psychology. (2000-2002)
Ph.D. University of North Carolina at Chapel Hill. Psychology. 2000
M.A. University of North Carolina at Chapel Hill. Psychology. 1997
B.A. University of Kansas. Psychology. Honors. 1995

Professional Experience

2002 – Present Assistant Professor, Department of Psychology, Middlebury College.
2000 – 2002 Post-doctoral Research Associate, Department of Psychology, Carnegie Mellon University.
1995 – 2000 Teaching and Research Assistant, Department of Psychology, University of North Carolina at Chapel Hill.

Publications Related to Project

Arndt, J., & Gould, C.* (under review). The contribution of activation and monitoring processes to false recognition. *Memory*.

Arndt, J. (2006). Distinctive information and false recognition: The contribution of encoding and retrieval factors. *Journal of Memory and Language*, 54, 113-130.

Arndt, J., & Reder, L. M. (2003). The effect of distinctive visual information on false recognition. *Journal of Memory and Language*, 48, 1-15.

Arndt, J. & Reder, L. (2002). Word frequency and receiver operating characteristic curves in recognition memory: Evidence for a dual-process interpretation. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 28, 830-842.

Arndt, J. & Hirshman, E. (1998). True and false recognition in MINERVA2: Explanations of a global matching perspective. *Journal of Memory and Language*, 39, 371-391.

Other Publications

Arndt, J., Passannante, A., & Hirshman, E. (2004). The effect of midazolam on implicit and explicit memory in category exemplar production and category cued recall. *Memory*, 12, 158-173.

Hirshman, E., Passannante, A., & Arndt, J. (2001). Midazolam amnesia and conceptual processing in implicit memory. *Journal of Experimental Psychology: General*, 130, 453-465.

Hirshman, E. & Arndt, J. (1997). Discriminating alternative conceptions of false recognition: The cases of word concreteness and word frequency. *Journal of Experimental Psychology: Learning, Memory, and Cognition*, 23, 1306-1323.

Park, H., Arndt, J., & Reder, L. (in press). A contextual interference account of distinctiveness effects in recognition memory. *Memory & Cognition*.

Diana, R.A., Reder, L.M., Arndt, J., & Park, H. (in press). Models of recognition: A review of arguments in favor of a dual-process account. *Psychonomic Bulletin & Review*.

Selected Presentations

Arndt, J., Osborne, A.*, Stone, M.*, Carney, J.*, & Lee, K.* (2005). Does distinctive visual information enhance monitoring of false memories? Poster presented at the annual meeting of the Psychonomic Society, Toronto, ON.

Arndt, J. (2005). Perceptually-based memories of events that were never perceived. Invited talk given at Psychology Department Colloquium, Colby College, Waterville, ME.

Gould, C.* & Arndt, J. (2005). The effects of repetition, associates and retrieval speed on false memory. Poster presented at the annual meeting of the Eastern Psychological Association, Boston, MA.

Arndt, J., & Carney, J.* (2004). Distinctive visual information and false recognition: The role of encoding and retrieval factors. Poster presented at the annual meeting of the Psychonomic Society, Minneapolis, MN.

Arndt, J., & Lee, K.* (2004). The effect of blocking study list presentation on false recognition. Poster presented at the annual meeting of the American Psychological Society, Chicago, IL.

Synergistic Activities

- Supervised 18 research students (research assistants, independent study students and thesis students) since arriving at Middlebury in 2002.

- Co-PI for Middlebury Psychology NSF-REU program in Summer, 2005 (supervised 5 students personally and co-coordinated program for 11 summer research students)

Collaborators and Other Affiliations

Collaborators: Elliot Hirshman, Ph.D., George Washington University
 Lynne M. Reder, Ph.D., Carnegie Mellon University
 Julie Dumas, Ph.D., University of Vermont
 Anthony Passannante, M.D., University of North Carolina, Chapel Hill
 Heekyeong Park, Ph.D., Carnegie Mellon University
 Julia Fisher, Ph.D., Boston College
 Thomas Henthorn, M.D., University of Colorado Health Sciences

Graduate Advisor: Elliot Hirshman, Ph.D., George Washington University

Post-doctoral Advisor: Lynne M. Reder, Ph.D., Carnegie Mellon University

SUMMARY PROPOSAL BUDGET

YEAR 1

ORGANIZATION Middlebury College				FOR NSF USE ONLY					
PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR Jason Arndt				PROPOSAL NO.		DURATION (months)			
				Proposed		Granted			
AWARD NO.									
A. SENIOR PERSONNEL: PI/PI, Co-PI's, Faculty and Other Senior Associates (List each separately with title, A.7. show number in brackets)				NSF Funded Person-months		Funds Requested By proposer		Funds granted by NSF (if different)	
				CAL	ACAD	SUMR			
1. Jason Arndt - Assistant Professor				0.00	1.80	2.00	\$ 26,000		
2.									
3.									
4.									
5.									
6. (0) OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)				0.00	0.00	0.00	0		
7. (1) TOTAL SENIOR PERSONNEL (1 - 6)				0.00	1.80	2.00	26,000		
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)									
1. (0) POST DOCTORAL ASSOCIATES				0.00	0.00	0.00	0		
2. (0) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)				0.00	0.00	0.00	0		
3. (0) GRADUATE STUDENTS							0		
4. (3) UNDERGRADUATE STUDENTS							7,039		
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)							0		
6. (0) OTHER							0		
TOTAL SALARIES AND WAGES (A + B)							33,039		
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)							9,986		
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)							43,025		
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING \$5,000.)									
TOTAL EQUIPMENT							0		
E. TRAVEL 1. DOMESTIC (INCL. CANADA, MEXICO AND U.S. POSSESSIONS)							6,000		
2. FOREIGN							0		
F. PARTICIPANT SUPPORT COSTS									
1. STIPENDS \$ _____				0					
2. TRAVEL _____				0					
3. SUBSISTENCE _____				0					
4. OTHER _____				0					
TOTAL NUMBER OF PARTICIPANTS (0) TOTAL PARTICIPANT COSTS							0		
G. OTHER DIRECT COSTS									
1. MATERIALS AND SUPPLIES							0		
2. PUBLICATION COSTS/DOCUMENTATION/DISSEMINATION							300		
3. CONSULTANT SERVICES							0		
4. COMPUTER SERVICES							0		
5. SUBAWARDS							0		
6. OTHER							11,900		
TOTAL OTHER DIRECT COSTS							12,200		
H. TOTAL DIRECT COSTS (A THROUGH G)							61,225		
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE) Salaries and Wages (minus sabbatical) (Rate: 62.0000, Base: 20723)									
TOTAL INDIRECT COSTS (F&A)							12,848		
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)							74,073		
K. RESIDUAL FUNDS (IF FOR FURTHER SUPPORT OF CURRENT PROJECTS SEE GPG II.C.6.j.)							0		
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)							\$ 74,073	\$	
M. COST SHARING PROPOSED LEVEL \$ Not Shown				AGREED LEVEL IF DIFFERENT \$					
PI/PI NAME Jason Arndt				FOR NSF USE ONLY					
ORG. REP. NAME* Cindy Wemette				INDIRECT COST RATE VERIFICATION					
				Date Checked		Date Of Rate Sheet		Initials - ORG	

SUMMARY PROPOSAL BUDGET

YEAR 2

ORGANIZATION Middlebury College				FOR NSF USE ONLY			
PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR Jason Arndt				PROPOSAL NO.	DURATION (months)		
				AWARD NO.	Proposed	Granted	
A. SENIOR PERSONNEL: PI/PI, Co-PI's, Faculty and Other Senior Associates (List each separately with title, A.7. show number in brackets)				NSF Funded Person-months		Funds Requested By proposer	Funds granted by NSF (if different)
				CAL	ACAD	SUMR	
1. Jason Arndt - Assistant Professor				0.00	0.00	2.00	\$ 14,368
2.							
3.							
4.							
5.							
6. (0) OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)				0.00	0.00	0.00	0
7. (1) TOTAL SENIOR PERSONNEL (1 - 6)				0.00	0.00	2.00	14,368
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)							
1. (0) POST DOCTORAL ASSOCIATES				0.00	0.00	0.00	0
2. (0) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)				0.00	0.00	0.00	0
3. (0) GRADUATE STUDENTS							0
4. (3) UNDERGRADUATE STUDENTS							7,241
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)							0
6. (0) OTHER							0
TOTAL SALARIES AND WAGES (A + B)							21,609
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)							5,853
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)							27,462
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING \$5,000.)							
TOTAL EQUIPMENT							0
E. TRAVEL 1. DOMESTIC (INCL. CANADA, MEXICO AND U.S. POSSESSIONS)							6,000
2. FOREIGN							0
F. PARTICIPANT SUPPORT COSTS							
1. STIPENDS \$ _____				0			
2. TRAVEL _____				0			
3. SUBSISTENCE _____				0			
4. OTHER _____				0			
TOTAL NUMBER OF PARTICIPANTS (0) TOTAL PARTICIPANT COSTS							0
G. OTHER DIRECT COSTS							
1. MATERIALS AND SUPPLIES							0
2. PUBLICATION COSTS/DOCUMENTATION/DISEMINATION							300
3. CONSULTANT SERVICES							0
4. COMPUTER SERVICES							0
5. SUBAWARDS							0
6. OTHER							2,500
TOTAL OTHER DIRECT COSTS							2,800
H. TOTAL DIRECT COSTS (A THROUGH G)							36,262
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE)							
Salaries & Wages (Rate: 62.0000, Base: 21609)							
TOTAL INDIRECT COSTS (F&A)							13,398
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)							49,660
K. RESIDUAL FUNDS (IF FOR FURTHER SUPPORT OF CURRENT PROJECTS SEE GPG II.C.6.j.)							0
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)							\$ 49,660 \$
M. COST SHARING PROPOSED LEVEL \$ Not Shown				AGREED LEVEL IF DIFFERENT \$			
PI/PI NAME Jason Arndt				FOR NSF USE ONLY			
ORG. REP. NAME* Cindy Wemette				INDIRECT COST RATE VERIFICATION			
				Date Checked	Date Of Rate Sheet	Initials - ORG	

SUMMARY PROPOSAL BUDGET

YEAR 3

ORGANIZATION Middlebury College				FOR NSF USE ONLY			
PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR Jason Arndt				PROPOSAL NO.	DURATION (months)		
				AWARD NO.	Proposed	Granted	
A. SENIOR PERSONNEL: PI/PI, Co-PI's, Faculty and Other Senior Associates (List each separately with title, A.7. show number in brackets)				NSF Funded Person-months		Funds Requested By proposer	Funds granted by NSF (if different)
				CAL	ACAD	SUMR	
1. Jason Arndt - Assistant Professor				0.00	0.00	2.00	\$ 15,087
2.							
3.							
4.							
5.							
6. (0) OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)				0.00	0.00	0.00	0
7. (1) TOTAL SENIOR PERSONNEL (1 - 6)				0.00	0.00	2.00	15,087
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)							
1. (0) POST DOCTORAL ASSOCIATES				0.00	0.00	0.00	0
2. (0) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)				0.00	0.00	0.00	0
3. (0) GRADUATE STUDENTS							0
4. (3) UNDERGRADUATE STUDENTS							7,447
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)							0
6. (0) OTHER							0
TOTAL SALARIES AND WAGES (A + B)							22,534
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)							6,131
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)							28,665
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING \$5,000.)							
TOTAL EQUIPMENT							0
E. TRAVEL 1. DOMESTIC (INCL. CANADA, MEXICO AND U.S. POSSESSIONS)							6,000
2. FOREIGN							0
F. PARTICIPANT SUPPORT COSTS							
1. STIPENDS \$ _____				0			
2. TRAVEL _____				0			
3. SUBSISTENCE _____				0			
4. OTHER _____				0			
TOTAL NUMBER OF PARTICIPANTS (0) TOTAL PARTICIPANT COSTS							0
G. OTHER DIRECT COSTS							
1. MATERIALS AND SUPPLIES							0
2. PUBLICATION COSTS/DOCUMENTATION/DISEMINATION							300
3. CONSULTANT SERVICES							0
4. COMPUTER SERVICES							0
5. SUBAWARDS							0
6. OTHER							2,500
TOTAL OTHER DIRECT COSTS							2,800
H. TOTAL DIRECT COSTS (A THROUGH G)							37,465
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE) Salaries & Wages (Rate: 62.0000, Base: 22534)							
TOTAL INDIRECT COSTS (F&A)							13,971
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)							51,436
K. RESIDUAL FUNDS (IF FOR FURTHER SUPPORT OF CURRENT PROJECTS SEE GPG II.C.6.j.)							0
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)							\$ 51,436 \$
M. COST SHARING PROPOSED LEVEL \$ Not Shown				AGREED LEVEL IF DIFFERENT \$			
PI/PI NAME Jason Arndt				FOR NSF USE ONLY			
ORG. REP. NAME* Cindy Wemette				INDIRECT COST RATE VERIFICATION			
				Date Checked	Date Of Rate Sheet	Initials - ORG	

SUMMARY PROPOSAL BUDGET

Cumulative

ORGANIZATION Middlebury College				FOR NSF USE ONLY			
PRINCIPAL INVESTIGATOR / PROJECT DIRECTOR Jason Arndt				PROPOSAL NO.	DURATION (months)		
				AWARD NO.	Proposed	Granted	
A. SENIOR PERSONNEL: PI/PI, Co-PI's, Faculty and Other Senior Associates (List each separately with title, A.7. show number in brackets)				NSF Funded Person-months		Funds Requested By proposer	Funds granted by NSF (if different)
				CAL	ACAD	SUMR	
1. Jason Arndt - Assistant Professor				0.00	1.80	6.00	\$ 55,455
2.							
3.							
4.							
5.							
6. () OTHERS (LIST INDIVIDUALLY ON BUDGET JUSTIFICATION PAGE)				0.00	0.00	0.00	0
7. (1) TOTAL SENIOR PERSONNEL (1 - 6)				0.00	1.80	6.00	55,455
B. OTHER PERSONNEL (SHOW NUMBERS IN BRACKETS)							
1. (0) POST DOCTORAL ASSOCIATES				0.00	0.00	0.00	0
2. (0) OTHER PROFESSIONALS (TECHNICIAN, PROGRAMMER, ETC.)				0.00	0.00	0.00	0
3. (0) GRADUATE STUDENTS							0
4. (9) UNDERGRADUATE STUDENTS							21,727
5. (0) SECRETARIAL - CLERICAL (IF CHARGED DIRECTLY)							0
6. (0) OTHER							0
TOTAL SALARIES AND WAGES (A + B)							77,182
C. FRINGE BENEFITS (IF CHARGED AS DIRECT COSTS)							21,970
TOTAL SALARIES, WAGES AND FRINGE BENEFITS (A + B + C)							99,152
D. EQUIPMENT (LIST ITEM AND DOLLAR AMOUNT FOR EACH ITEM EXCEEDING \$5,000.)							
TOTAL EQUIPMENT							0
E. TRAVEL 1. DOMESTIC (INCL. CANADA, MEXICO AND U.S. POSSESSIONS)							18,000
2. FOREIGN							0
F. PARTICIPANT SUPPORT COSTS							
1. STIPENDS \$ _____				0			
2. TRAVEL _____				0			
3. SUBSISTENCE _____				0			
4. OTHER _____				0			
TOTAL NUMBER OF PARTICIPANTS (0) TOTAL PARTICIPANT COSTS							0
G. OTHER DIRECT COSTS							
1. MATERIALS AND SUPPLIES							0
2. PUBLICATION COSTS/DOCUMENTATION/DISEMINATION							900
3. CONSULTANT SERVICES							0
4. COMPUTER SERVICES							0
5. SUBAWARDS							0
6. OTHER							16,900
TOTAL OTHER DIRECT COSTS							17,800
H. TOTAL DIRECT COSTS (A THROUGH G)							134,952
I. INDIRECT COSTS (F&A)(SPECIFY RATE AND BASE)							
TOTAL INDIRECT COSTS (F&A)							40,217
J. TOTAL DIRECT AND INDIRECT COSTS (H + I)							175,169
K. RESIDUAL FUNDS (IF FOR FURTHER SUPPORT OF CURRENT PROJECTS SEE GPG II.C.6.j.)							0
L. AMOUNT OF THIS REQUEST (J) OR (J MINUS K)							\$ 175,169 \$
M. COST SHARING PROPOSED LEVEL \$ Not Shown				AGREED LEVEL IF DIFFERENT \$			
PI/PI NAME Jason Arndt				FOR NSF USE ONLY			
ORG. REP. NAME* Cindy Wemette				INDIRECT COST RATE VERIFICATION			
				Date Checked	Date Of Rate Sheet	Initials - ORG	

C *ELECTRONIC SIGNATURES REQUIRED FOR REVISED BUDGET

Budget Justification

PERSONNEL

PI Salary

Two months of summer salary support for the PI are requested per year of the grant period. The PI is eligible for up to 80% salary during his sabbatical leave year (2006-2007). NSF funding is requested for 20% salary to enable him to work full time on this project. As an indication of its commitment to research, Middlebury College has agreed to waive Facilities & Administration costs (Indirect Costs) on the PI's 2006-2007 regular year sabbatical funding. The PI will be the primary generator of research studies and dissemination of the results of research studies, either via directly designing and writing about studies, or via mentoring research students pursuing independent research. He will also recruit and mentor research assistants during each academic year and during the summers of 2007, 2008, and 2009. In addition to supporting the PI during the time that he mentors summer research students, summer salary will support the PI while he is writing research reports for disseminating the results of the proposed studies. Given the rigors of the semester at Middlebury, the PI tends to focus his research efforts on data collection (and the educational elements for students that accompany data collection) during the academic year and writes papers describing the results of that research during the summer months. Data collection is accomplished during the summer months when possible (i.e., when funds are available to recruit paid participants). However, given that Middlebury does not hold regular classes during the summer months, the number of available research participants is limited to the approximately 100 summer research students that are on campus each year.

Student Research Assistants

Funds to support two undergraduate research assistants per academic year (10 hours per week per student) and one undergraduate research assistant per summer (full time – 40 hours per week) are requested. These funds will provide wages to pay student research assistants at the current rates paid by Middlebury College to student workers. Student research assistant duties will involve all elements of the research process: study conceptualization, study design, stimulus construction, experiment programming, subject recruitment, data collection, data scoring, data management, data analysis, and results interpretation. Student research assistants will meet with the PI regularly to discuss readings relevant to the studies being conducted (as well as how the ideas in those readings are related to the studies being conducted), the progress of data collection, and the direction in which additional follow-up studies will be conducted. Finally, provided their interest, research assistants will be involved in the dissemination process, either via construction of research posters for presentation at professional conferences, or via writing portions of research reports describing the results of the studies described in this proposal.

Reviewers of this grant may wonder why funding for only one summer research student per year is requested, since summers tend to be the most productive research time for faculty at institutions where one's primary devotion during the semester is teaching. At Middlebury, there tend to be a limited number of participants available during the summer months because Middlebury does not hold regular classes during the summer months, because Middlebury's Language Schools are in session then. This past summer as part of our REU program, my research students and I tried running a single set of subjects through a series of studies along the lines of those described in this proposal. This exploration was, in effect, an examination of whether it was possible to utilize the same set of subjects in a series of related studies, information that would be useful to me for planning subsequent summer research. What we found was that as participants progressed through the summer, they seemed to change the strategic processes that they utilized. Subjects tended to become exceedingly conservative in responding on recognition memory tests, presumably in an effort to avoid committing errors. While such a result may be interesting

in its own right, when one is trying to examine other memory processes, it becomes a substantial complication. Thus, I believe that there are probably insufficient participants on campus during the summer for more than one research student to utilize in her or his research.

Finally, in addition to mentoring the students funded by this grant, the PI will engage additional Middlebury students in research beyond those funded by this grant via senior thesis mentoring, independent study mentoring, and directed research mentoring, all of which are completed by students for academic credit. Further, the PI may hire some students using either or both of Middlebury's internal funds devoted to funding faculty-student collaboration on research, the Undergraduate Collaborative Research Fund and the Faculty Research Assistant Fund.

Fringe Benefits: Fringe benefits are requested at the Middlebury College negotiated rate of 35.7% for faculty salary and at 10.0% for student stipends/wages.

TRAVEL

Two essential parts of fully exposing students to the research process is providing them with experience disseminating the results of the studies they have conducted and talking with other researchers about their work. The environment most conducive to student learning in this regard is the academic conference. For each year of the grant period, \$6000 is requested for travel to conferences for the PI and students (\$18000 total over three years). This amount allows \$1500 for travel to two conferences per year for the PI, and \$1500 for each of two students to accompany the PI to one conference each year.

OTHER DIRECT COSTS

Participant Payment

\$2000 per year of the grant period (\$6000 total) is requested to pay participants for participation in the research studies outlined in the proposal. These funds will allow for the collection of approximately 200 participants per year, each paid \$10 per session for their participation. These funds will primarily be used during the summer months, when Middlebury College does not hold regular classes (Middlebury has an excellent language school during the summers, and students on campus for the language schools pledge to not speak English during their studies). As a result, during the summer months, a Psychology Department subject pool is unavailable for recruiting participants. Thus, payment of participants is necessary in order for summer research students to collect data. These funds may also be used during the regular school year to supplement the number of participants available via the Psychology Department subject pool.

Publication Costs and Dissemination

\$300 per year of the grant period is requested to fund publication and dissemination of the results of these studies (\$900 total). These funds will primarily go toward ordering reprints of the publications that result from the studies that will be conducted as part of the research program outlined in this grant proposal. In addition, these funds will contribute to the costs of printing research posters for presentation at conferences where the PI and the students funded by this grant present their research.

Software

Funding to purchase a 5 license version of e-prime v2.0 is requested (\$3000). E-prime serves as the primary experiment-generation platform that will be used to program the experiments outlined in this grant application. E-prime is ideal for the involvement of undergraduates in research because it utilizes a point and click interface that is easy to learn for novice programmers. This feature allows research

assistants to become accomplished programmers and learn to construct even sophisticated, complex experiments relatively early in their internship with the PI. Multiple licenses are requested to allow multiple students to program experiments on the lab machines in the PIs lab, as well as for the PI to program on his office computer when necessary.

Hardware

Funding to purchase four (4) new Dell PCs is requested for data collection (\$1600 each, \$6400 total). These four PCs will replace the PIs current machines, which are nearing the end of their period as reliable working machines – all will be four years old in June, 2006. These four PCs will serve multiple functions toward completion of the proposed research. First, each of the four PCs will be outfitted with e-prime v2.0 software (see above section), which will allow students to both learn to use e-prime and program experiments that will be run to complete the proposed series of research studies. Second, these four PCs will serve as data collection terminals for research assistants to use to run the studies outlined in this proposal. Third, the PCs will serve as work stations where research students can score, compile, and analyze data that are collected. Fourth, research students will have access to lab machines to work on posters that will be presented at conferences in order to disseminate the results of our studies and to expose research students to the conference environment. Finally, work stations can be used by research students to write portions of research reports that will be submitted to scholarly journals to disseminate the results of these studies to the research community.

Weekly Lab Meeting Lunch

As outlined in the RUI Impact Statement, I hold lab meetings each Friday when classes are in session. These lab meetings serve multiple purposes academically for the students. As was noted in the Impact Statement, these lab meeting offer students the opportunity to present their projects and discuss papers from the research literature that are related to those projects. Further, they allow students to be kept abreast of all of the different projects that are ongoing in the lab at any give time. Importantly, they also offer an opportunity for informal inspiration of more novice research students by more advanced research students.

In addition to these significant academic benefits, I feel that lab meetings also offer numerous advantages in building cohesion, cooperation, and rapport among the lab students. It is possible, given the relatively solitary nature of cognition research (one experimenter at a time in the lab, usually running a single participant, and often working on individual projects where I am their primary collaborator), for lab students to rarely interact with one another. Weekly lab meetings offer the students an opportunity to get to know one another and to develop a sense of a lab community that is otherwise unlikely to arise spontaneously given the limited interaction they would have studying cognition. In order to enhance this feeling of community, and to allow the students an opportunity to meet in a more informal atmosphere (i.e., not in the lab and not in a campus classroom), we hold lab meetings over lunch at the campus eatery, *The Grille* (where I pick up the tab). The students enjoy meeting there, because it is a break in their lunch routine of going to a dining hall. Further, I feel that the relatively informal nature of the environment at *The Grille* allows students to see that research is not something that only occurs within the confines of a laboratory, or at prescribed times during meetings in my office. Rather, it shows them that research is something that a person comes to love, to the point that it is fun and exciting to “talk shop” informally over a meal. Indeed, I believe that in many ways these meetings closely approximate the way many real research discussions occur among professionals – over a lunch meeting when two (or more) researchers simply sit down to think and freely exchange ideas. Thus, I have included a request in the budget to fund these weekly lunch meetings between myself and the students working in my lab. The typical cost of lunch for myself and the 4-5 students is approximately \$25. I have budgeted funds for 20 lab meetings per year at \$25 each (\$500 per year, \$1500 total).

(See GPG Section II.C.2.h for guidance on information to include on this form.)

Investigator: **Jason Arndt**

Support: ☐ Current ☒ Pending ☐ Submission Planned in Near Future ☐ *Transfer of Support

Project/Proposal Title: **Associations between perceptual information and unstudied information**

Total Award Amount: \$ **197,850** Total Award Period Covered: **08/01/06 - 07/31/09**

Person-Months Per Year Committed to the Project. Cal:0.00 Acad: 1.00 Summr: 2.00

Support: ☐ Current ☒ Pending ☐ Submission Planned in Near Future ☐ *Transfer of Support

Project/Proposal Title: **The contribution of repetition and source variability to political persuasion**

Total Award Amount: \$ **25,000** Total Award Period Covered: **09/01/06 - 08/31/07**

Person-Months Per Year Committed to the Project. Cal:0.00 Acad:4.00 Sumr: 0.00

Support: ☒ Current ☐ Pending ☐ Submission Planned in Near Future ☐ *Transfer of Support

Project/Proposal Title: **Research in Psychology: Discovering the Process**

Total Award Amount: \$ **60,032** Total Award Period Covered: **03/18/05 - 03/15/06**

Person-Months Per Year Committed to the Project.	Cal:0.00	Acad: 0.00	Sumr: 2.00
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Support: ☐ Current ☒ Pending ☐ Submission Planned in Near Future ☐ *Transfer of Support

Project/Proposal Title: **RUI: Memory for perceiving information that was never experienced**

Total Award Amount: \$ **175,169** Total Award Period Covered: **09/01/06 - 08/31/09**

Person-Months Per Year Committed to the Project. Cal:0.00 Acad: 0.60 Sumr: 2.00

Support: ☐ Current ☐ Pending ☐ Submission Planned in Near Future ☐ *Transfer of Support

Project/Proposal Title:

Total Award Amount: \$ Total Award Period Covered:

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USE ADDITIONAL SHEETS AS NECESSARY

FACILITIES, EQUIPMENT & OTHER RESOURCES

FACILITIES: Identify the facilities to be used at each performance site listed and, as appropriate, indicate their capacities, pertinent capabilities, relative proximity, and extent of availability to the project. Use "Other" to describe the facilities at any other performance sites listed and at sites for field studies. USE additional pages as necessary.

Laboratory: **The PI has a research laboratory dedicated to his research. The lab has four (4) Dell PCs running e-prime experiment generation software for experiment generation and running, as well as a response box for precise response time measurement, and a microphone for vocal response recording.**

Clinical: **N/A**

Animal: **N/A**

Computer: **Middlebury College maintains excellent computer labs available to all students, with software needed for all forms of data management and analysis (Excel and SPSS), word processing (Word), as well as poster and presentation production (Adobe Illustrator and Power Point). In the**

Office: **Clerical support will be provided by Middlebury College.**

Other:

MAJOR EQUIPMENT: List the most important items available for this project and, as appropriate identifying the location and pertinent capabilities of each.

OTHER RESOURCES: Provide any information describing the other resources available for the project. Identify support services such as consultant, secretarial, machine shop, and electronics shop, and the extent to which they will be available for the project. Include an explanation of any consortium/contractual arrangements with other organizations.

FACILITIES, EQUIPMENT & OTHER RESOURCES

Continuation Page:

LABORATORY FACILITIES (continued):

The lab computers also have Microsoft Office that can be used to score and compile data, as well as for word processing or poster and presentation production.

COMPUTER FACILITIES (continued):

building where the Psychology Department is located, there are three (3) computer laboratories available for student use. Around campus, there are numerous other laboratories available for students use, all of which contain most or all of the software noted (MS office, SPSS, Adobe Illustrator).

Impact of the Proposed Research on Undergraduate Education

Middlebury College and the Psychology Department

Middlebury College is a private four-year college founded in 1800 that is located in Middlebury, Vermont. Middlebury is regularly listed among the top liberal arts colleges in the nation in the *US News and World Report* rankings as well as other annual listings. The student body consists of approximately 2350-2400 students seeking degrees in the social and natural sciences, humanities, foreign languages, and the arts. The college's student population currently comes from all 50 states, and 68 foreign countries. International students comprise 8.1% of Middlebury's student body. Additionally, of the US students at Middlebury, approximately 19% are students of color (8% Asian American, 6% Hispanic, 3% African American, and 3% Native American or "Other"). Thus, considering its location in rural Vermont, Middlebury has an exceptionally diverse student body.

Middlebury College has a deep institutional commitment to supporting research collaboration between students and faculty. The teaching load (2 courses per semester) and relatively small class size is structured to afford faculty time to engage undergraduates in research. The College has internal funding mechanisms (the Undergraduate Collaborative Research Fund and the Faculty Research Assistant Fund) that provide support for faculty-student research collaboration. The College provides funding for independent work conducted by seniors in the amount of \$350 for every applicant. In addition, seniors can apply to a competitive fund that provides up to \$1500 for senior independent research. This institutional commitment to student engagement in research has been successful in inspiring students to pursue graduate study. Alumni records indicate that 112 Middlebury graduates have received an academic or professional doctorate in the last five years. Finally, the College has a fund specifically devoted to faculty professional development, from which each faculty member can use up to \$2000 annually. This fund may be used for any activity related to professional development, such as conference attendance or research expenses.

The Psychology Department is housed in Middlebury's \$47 million science building (built in 1999), McCardell Bicentennial Hall. The building contains extensive lab space for all disciplines housed in it (Psychology, Physics, Computer Science, Chemistry, Geology, Biology, and Geography), and is among the most impressive science buildings at an institution of Middlebury's size. McCardell Bicentennial Hall was awarded R & D Magazine's "Lab of the Year" for 2000. Psychology students have access to many types of instrumentation needed to conduct top-quality Psychology research, including video-wired observation rooms, computer laboratories, skin conductance equipment, and animal testing facilities.

The Psychology Department currently has 11 full-time faculty, all of whom have Ph.D.'s. The Psychology major is built around a curriculum where students are educated in four core areas (Cognition and Neuroscience, Social and Applied, Clinical and Personality, and Developmental). On average, more than 30 Psychology majors a year participate in either directed research or a one-semester independent study in collaboration with a faculty member. In addition, the Psychology Department is currently mentoring the largest number of senior thesis students in its history, with 23 students on track to complete a senior honors thesis this academic calendar year. Historically, approximately 15% of the 60-70 graduating Psychology majors pursue a senior thesis each year. Over the last 10 years, the department faculty have had over 80 student co-authors on papers and conference presentations. Thus, the general atmosphere both at

Middlebury College and in the Psychology Department is conducive to research productivity in general, and faculty-student research collaboration in particular.

The PI's qualifications

In my relatively short career at Middlebury (3.5 years), I have mentored 18 research students as directed research students (during the academic year or during the summer), independent study students, or senior thesis students. One of those students is in her first year in the Neuroscience Ph.D. program at Duke University, one will be entering graduate study in Linguistics at Indiana University beginning in 2006, and two students are currently applying to Ph.D. programs in Psychology and Neuroscience. Further, I know of at least seven other students who have either worked in my lab or are currently working in my lab who plan to apply to science Ph.D. programs in the coming few years. In addition, I was one of two coordinators for the Psychology Department's Research Experience for Undergraduates (REU) program in Summer, 2005 funded by NSF. As part of this program, I personally mentored five research interns, and co-coordinated the program for all 11 students.

On four occasions over the past two years, I have presented research in conjunction with undergraduate co-authors at regional or national conferences, with seven student co-authors on those conference presentations. In addition, two more abstracts (one poster and one paper) have been submitted to the 2006 Eastern Psychological Association meeting, with five student co-authors on the poster and six student co-authors on the paper. These presentations are the result of research conducted this past summer as part of the Research Experience for Undergraduates program noted above. Because multiple students worked intensively to ensure the success of each project, I deemed all to be worthy of authorship on these presentations. I have further encouraged all of these students to attend the conference, and all five intend to join me. Finally, I recently co-authored a manuscript with a former thesis student (Christine Gould) that reports her thesis experiment as well as a second study designed to complement it. This manuscript is in the process of being revised after receiving a positive set of reviews of the initial submission. None of the reviewers suggested there were any "critical" issues that would preclude publication, thus I believe the paper stands a reasonable chance of being accepted upon submission of a revised version of the manuscript.

While the intent of the above statistics is to provide evidence that I regularly involve students in research, I feel it is also appropriate here to discuss the education received by all students who conduct research under my direction. I generally begin any student's tenure in my lab by assigning them several readings that provide them with the relevant theoretical background for understanding the study they are running. After the student has had an opportunity to read the papers, we meet individually (or with any other students also working on that project) to discuss the theoretical ideas that led to the study they will be conducting. As part of these discussions, I strive to not only help the students appreciate the theories that explain the memory phenomena we are investigating, but also help them understand how our study tests those theories. After some of the data have been collected, I meet with the students to discuss data scoring, where I educate them in how to use Excel to score each participant's data, as well as how to set up a summary data sheet in Excel. We then meet periodically to discuss the results of the study at different stages of data collection, and what those data suggest about the theories and hypotheses our study was designed to test.

Following the collection of the data for a study or series of studies, students receive experience in several aspects of dissemination through my normal lab structure. The first experience is in the form of informal presentations in weekly lab meetings, the nature of which will be described in greater detail below. Presentations in these informal meetings allow students to become comfortable discussing and presenting their results. The informal atmosphere allows me to provide them with direction and feedback on their presentation style as well as any interpretations of their data that they provide but to do so in an environment where students feel more relaxed than at a conference or other public presentation venue. The second type of dissemination in which I involve students is the process of manuscript writing. Depending on the student's level of involvement in the project (thesis student, independent study student, directed research student) and interest, the student will play a larger or smaller role in writing up the results of the research. I offer thesis students the opportunity to write a paper describing their thesis study (with my help), which carries with it first author privilege. Independent study students, who typically have a less autonomous role in conceptualizing and conducting their own studies than thesis students, will be asked to participate in writing a manuscript, either at the level of writing up the methodology section and results, or at the level of theoretical discourse, depending on their interest level. Directed research students, provided their interest, will be asked to write up methodology and results sections, but typically will not be requested to contribute to writing the bulk of the manuscript. However, both independent study students and directed research students will be involved in the writing process by reading drafts of papers that I write, and being asked to comment on them. This will allow them to see their projects to fruition in the form of a manuscript submitted to a scholarly journal for publication. The third manner in which I involve students in dissemination is via conference attendance. Any time a student and I submit a poster or paper proposal to a conference, I invite students who have worked on that project to attend the conference with me. Students and I typically submit poster applications to conferences rather than talk applications because I believe that poster presentations allow a much better environment for students to interact with other researchers regarding their work. Finally, although it does not offer the students a direct view of the dissemination of the research projects they worked on, when I receive reviews of a manuscript, I share those with all of my lab students, along with the draft of the paper that generated the reviews. We then use one of our weekly lab meetings to discuss the paper as well as the reviews and how the reviewers' comments can be addressed in a revision of the manuscript or with further experimental work. This allows the students to gain experience with some elements of the publication process, regardless of whether or not they ultimately choose to participate heavily in the dissemination of their own research.

The final pedagogical instrument I use to educate students about research is that I ask all of my research students to attend a lab meeting over lunch each Friday. In these lab meetings, students are asked to make informal presentations about the projects they are working on. In preparation for each lab meeting, the presenters and I agree on a paper for the other lab students to read. This paper is chosen to serve as background for the research project that will be presented that week. During lab meetings, I generally assist the presenters in both discussing the background article that was read that week, as well as how it is related to the project they are presenting. However, I do my best to allow the students to be autonomous in presenting the data in an effort to foster ownership over the project and a deeper sense of commitment to the research.

In addition to giving each student the opportunity to both describe his or her work to others, weekly lab meetings allow interaction between students who are working on projects at different levels (thesis, independent study, directed research). I have found in the past two years of holding lab meetings that one of the most gratifying effects of the student-student interactions is that the more advanced students (thesis, independent study) can inspire directed research students to pursue their own research by presenting their projects to the lab. This was first made apparent to me in the Spring, 2005 semester when one of my directed research students came to me and asked if she could pursue an independent research project. Her motivation was the end of the semester presentation given by an independent study student the prior fall in lab meeting. She thought that designing her own research project would be fun and rewarding. We jointly read a series of papers related to the interests she expressed (social influences on memory), met extensively over the next couple of weeks, and designed a study. She then took to all of the hard work of putting together the stimuli and procedure, and ran the study, all within the confines of a 12 week semester. It was gratifying to see a student become so passionate about research, and an important lesson about the inspiration that students derive from seeing their peers perform novel research! I was treated to a similar experience this past fall. In casual conversation during our lab meetings, students often inquire about different elements of being a professor and being an academic. Among the questions during one conversation was about the publication process, and how often it was that student co-authored papers with me. I mentioned that a former thesis student and I had recently submitted a paper to a journal that described her thesis and a follow-up study that we had conducted. One of the students was very excited at this prospect and expressed the sentiment of “wow, that would be so *cool*.” Indeed, I believe that experiences like these, borne out of interactions between more advanced research students and more novice research students can be among the most valuable and motivating in bringing young researchers to the point where they wish to pursue science independently.

Specific impact of this project on undergraduate education

All aspects of the proposed research program are amenable to extensive student involvement, as I have documented above. Student researchers will be involved in stimulus construction, experiment programming, data collection, data scoring and data analysis for all of the studies outlined in this proposal. Thus, they will receive intimate familiarity with how to conduct research. Further, student researchers will gain experience in the conceptual elements of conducting research in two primary ways. First, the students will be extensively educated in the background for each of the research projects that are conducted as part of the proposed line of research. This education will be conducted by having directed research students read background research papers that are relevant to this line of studies and by engaging them in discussion of those articles in individual meetings or within the context of a lab meeting. Independent research students will gain knowledge of the background of a research study they conduct by reading the background literature with the goal of designing their own research study. This process will be performed in collaboration with myself, and I will act as a “reviewer” of their projects to ensure that the studies students design both contribute to the research literature in a meaningful way and are methodologically sound. Second, students will be involved in the dissemination process to a level that is appropriate to their role in the research. Regardless of whether the students take an active role in writing conceptual portions of a paper (i.e., not simply methodology and results sections), they will be asked to read manuscripts written by myself and more advanced students,

which we will then discuss in lab meetings. Further, upon receipt of the reviews of a manuscript, all students who worked on that project will be sent a copy of the reviews, and will be engaged in discussion (via email or in person, provided they are still on campus as a student) regarding the reviewers' comments and how the paper can be modified to address them. If a student played an integral role in initially writing the manuscript, we will then begin the process of jointly working on the revision of the manuscript, either via re-writing or via additional empirical work. Thus, students working on the research described in this proposal will gain intensive, first-hand experience in all elements of conducting research, from initial conceptualization to dissemination.

While this grant specifically provides funds for three students per year (two academic year students and one summer student), its impact will be far greater than that, as I anticipate mentoring students beyond those specifically funded by this grant. In particular, it is likely that thesis students and independent research students that are not paid salary from this grant will elect to conduct studies that fall within the framework of those outlined in this proposal. Thus, more than the nine students who are directly funded by this grant will benefit from it by receiving first-hand research experience. In addition, the students who are funded directly by this grant will enter a research environment where they will be strongly encouraged to progress up the ladder of autonomy and take on additional responsibilities in conducting research, to the point that they ultimately become independent researchers. Thus, one of the primary goals of engaging students in research stemming from this proposal is to give them preliminary experience in conducting research, which they can then use as they become increasingly autonomous and begin designing and executing their own research.

Further, as mentioned above, integrating students in to a research lab has multiple benefits, such as the inspiration more novice student researchers gain from hearing about the projects being conducted by more senior, independent students. This, I anticipate, will have the effect of motivating students who enter the lab as directed research students to pursue their own research ideas and projects, whether in my laboratory or in the labs of other faculty members. In addition to generally exposing a number of students to research, this grant provides funding for students to travel to professional conferences. This will have substantial impact on the students who work on research funded by this grant, as they will see first hand the excitement that comes with presenting one's results at a conference, as well as learning about the work conducted by other researchers that are in attendance at the conference.

Finally, the impact of involving undergraduates in research funded by this grant will be measured in four primary ways, most of which were described above when discussing my history mentoring undergraduates in research. First, I will assess the extent to which I am successful inspiring students to pursue independent research projects. This will be assessed by tracking the number of students who begin as research assistants who later elect to pursue independent research, either under my direction or under the direction of a faculty colleague. Second, I will keep track of the number of students who co-author conference presentation and publications with me. Third, I will track the number of students who attend conferences with me to present their research. Fourth, and finally, I will track the number of students who collaborate with me on research who later elect to attend graduate school. I believe these four statistics will provide a relatively comprehensive picture of how effectively undergraduate engagement in research encourages those same students to later pursue careers in science.

-----**Certification of RUI Eligibility**-----

By submission of this proposal, the institution hereby certifies that the originating and managing institution is an institution that offers courses leading to a bachelor's or master's degree, but has awarded an average of no more than 10 doctoral degrees per year in NSF-supported disciplines over the 2-to-5-year period preceding proposal submission.

Authorized Institutional Representative

Carol E. Rifelj
Dean for Faculty Development and Research

Typed Name and Title

<u>Carol Rifelj</u>	<u>8/31/05</u>
Signature	Date